

CHAPTER ONE

Data Communication And Computer Networks

Data Communication Basics

Communication Basics

- ❖ **Data communications** are the exchange of data between two devices via some form of transmission medium such as a wire cable.
- ❖ It includes :
 - Writing and talking
 - Nonverbal communication (such as facial expressions, body language or gestures)
 - Visual communication (the use of images or pictures such as painting, photography, video or film)
 - Electronic Communication such as telephone calls, electronic mail, cable television, satellite broadcasts.

Communication Basics

- ❖ For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs).

Five components of data communication

1. **Message:** is the information (data) to be communicated. Popular forms of information include **text, numbers, pictures, audio, and video.**
2. **Sender:** is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.
3. **Receiver:** is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.

Five components of data communication

3. **Transmission medium:** is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiberoptic cable, and radio waves.
4. **Protocol:** is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

Modes of data transmission

- ❖ There are 3 different transmission modes characterized according to the direction of the exchanges:
 1. **A simplex connection** is a connection in which the data flows in only **one direction**, from the transmitter to the receiver.
 - This type of connection is useful if the data do not need to flow in both directions (for example, from your computer to the printer or from the mouse to your computer...).

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2. **A half-duplex connection** (sometimes called an *alternating connection or semi-duplex*) is a connection in which the data flows in one direction or the other, but not both at the same time.
- With this type of connection, each end of the connection transmits in turn.
 - This type of connection makes it possible to have bidirectional communications using the full capacity of the line.

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- In a half-duplex transmission, the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time.
- Walkie-talkies and CB (citizens band) radios are both half-duplex systems.
- The half-duplex mode is used in cases where there is no need for communication in both directions at the same time; the entire capacity of the channel can be utilized for each direction.

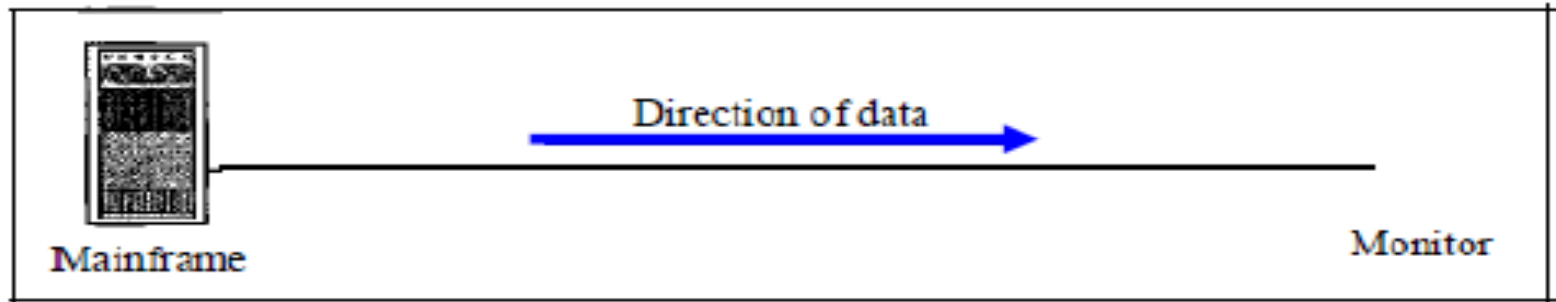
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- 3. Full-Duplex:** In full-duplex mode (also called duplex), both stations can transmit and receive simultaneously
- The full-duplex mode is like a two-way street with traffic flowing in both directions at the same time.
 - In full-duplex mode, signals going in one direction share the capacity of the link: with signals going in the other direction.
 - The full-duplex mode is used when communication in both directions is required all the time.
 - The capacity of the channel, however, must be divided between the two directions.

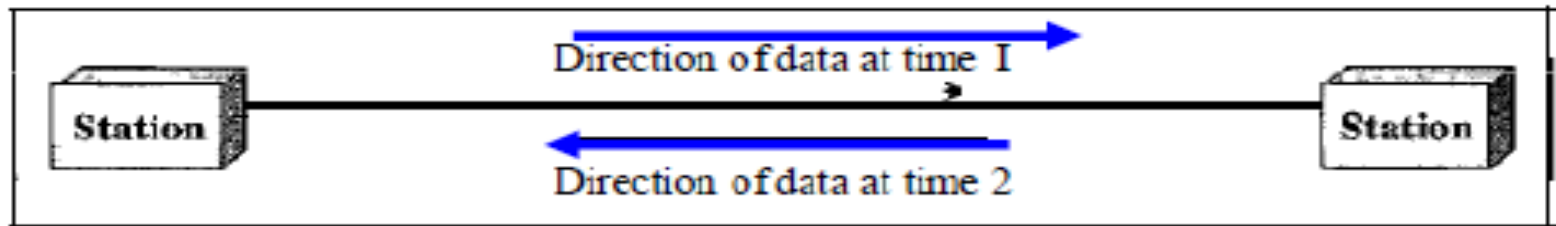
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- This sharing can occur in two ways: Either the link must contain two physically separate transmission paths, one for sending and the other for receiving; or the capacity of the channel is divided between signals travelling in both directions.
- One common example of full-duplex communication is the telephone network. When two people are communicating by a telephone line, both can talk and listen at the same time.

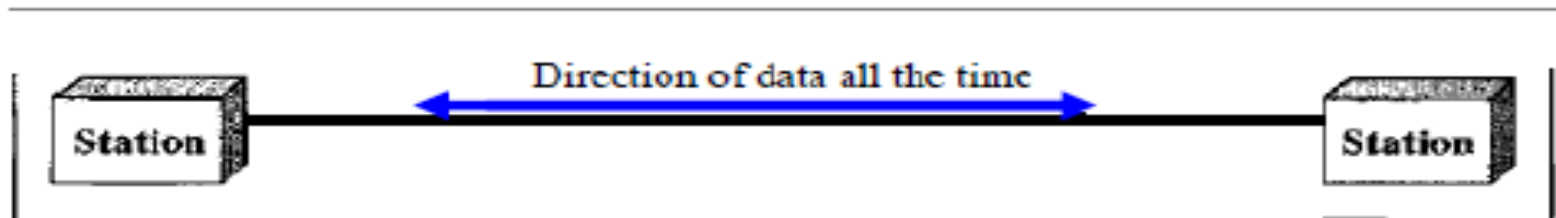
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a. Simplex



b. Half-duplex

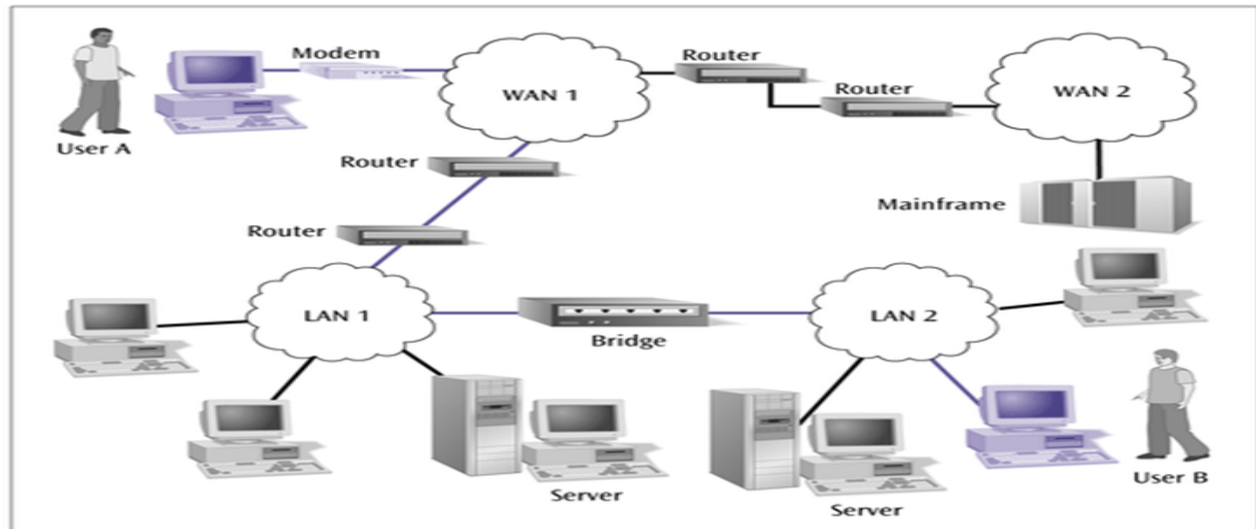


c. Full-duplex

Interconnection of Networks: Internetwork

- Today, it is very rare to see a LAN, a MAN, or a WAN in isolation; they are connected to one another.
- When two or more networks are connected, they become an internetwork, or **internet**.

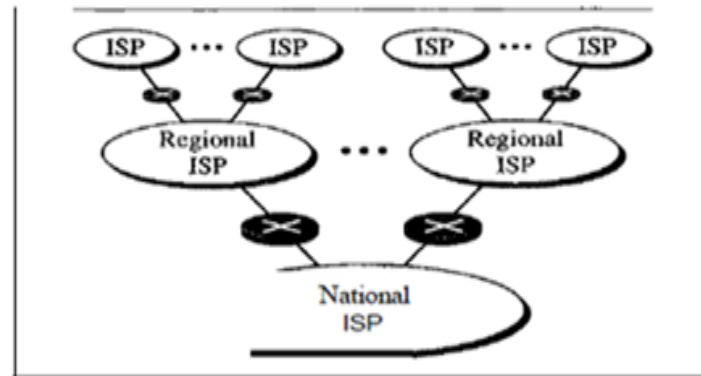
An overall view of the interconnection between local area networks and wide area networks



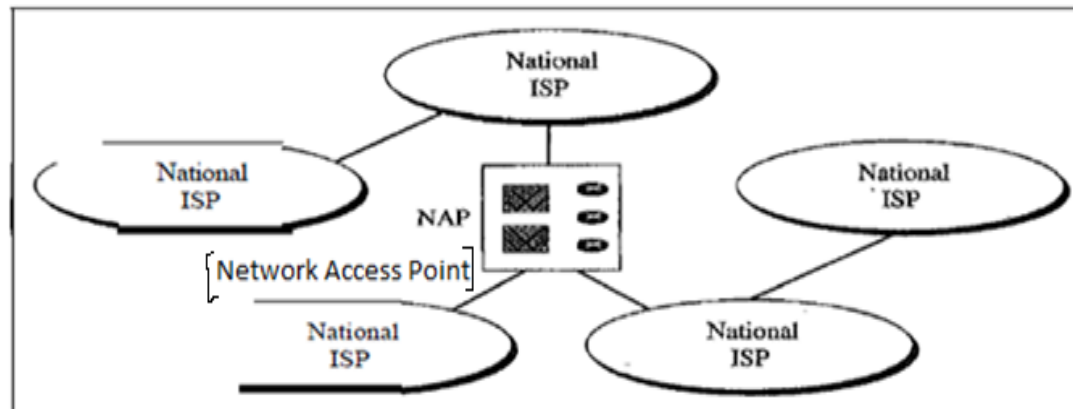
Network, internet, and Internet

- A **network** is a group of connected communicating devices such as computers and printers.
- An **internet** (note the lowercase letter i) is two or more networks that can communicate with each other.
- The most notable internet is called the **Internet** (uppercase letter I), a collaboration of more than hundreds of thousands of interconnected networks.
- Private individuals as well as various organizations such as government agencies, schools, research facilities, corporations, and libraries in more than 100 countries use the Internet.

Hierarchical organization of the Internet



a. Structure of a national ISP



b. Interconnection of national ISPs

Internet Vs. Intranet Vs. Extranet

- **Internet:** *It is a global collection of networks, both big and small. It is a giant network.*
- **Intranet:**
 - Are similar to World Wide Web (WWW) where centrally stored hypertext documents can be accessed using the web.
 - Unlike the WWW, intranets are **available only to user within the company network system**. Intranet is a very common service in large networked organizations.

Internet Vs. Intranet Vs. Extranet

- **Extranet:**

- Although similar to intranet, extranet provides selected users from outside the organization to access data from the internal network.
- Extranets are commonly used by suppliers to provide data to company clients.

Computer Network and its Applications

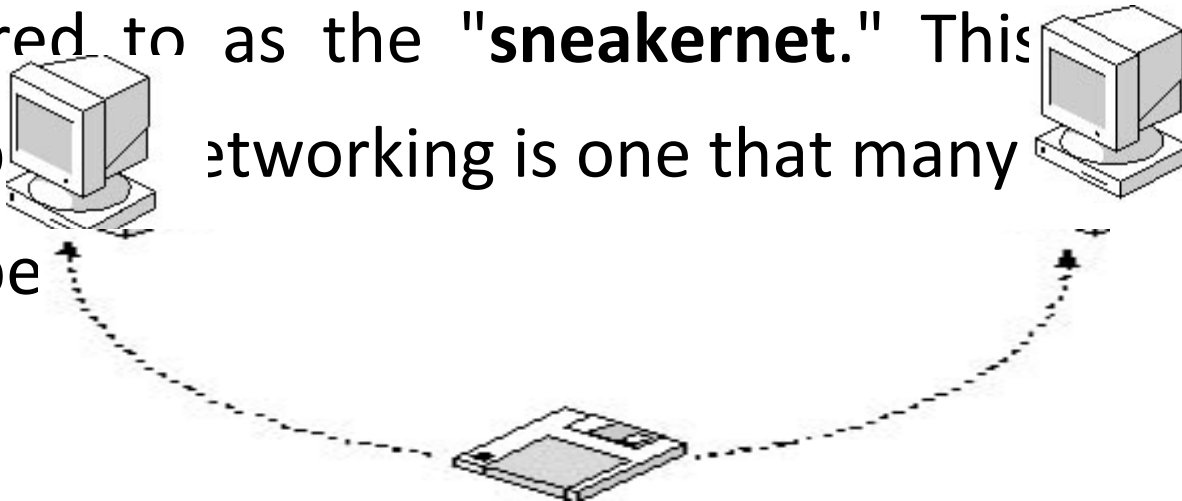
- 📢 At its most elementary level, a computer network consists of **two computers** connected to each other by a cable that allows them to share data.
- 📢 All computer networking, no matter how sophisticated, stems from that simple system.
- 📢 Computer networking arose as an answer to the need to **share data** in a timely fashion.
- 📢 Personal computers are powerful tools that can process and manipulate large amounts of data quickly, but they do not allow users to share that data efficiently.
- 📢 Before networks, users needed either to print out documents or copy document files to a disk for others to edit or use them.

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📢 If others made changes to the document, there was no easy way to merge the changes.

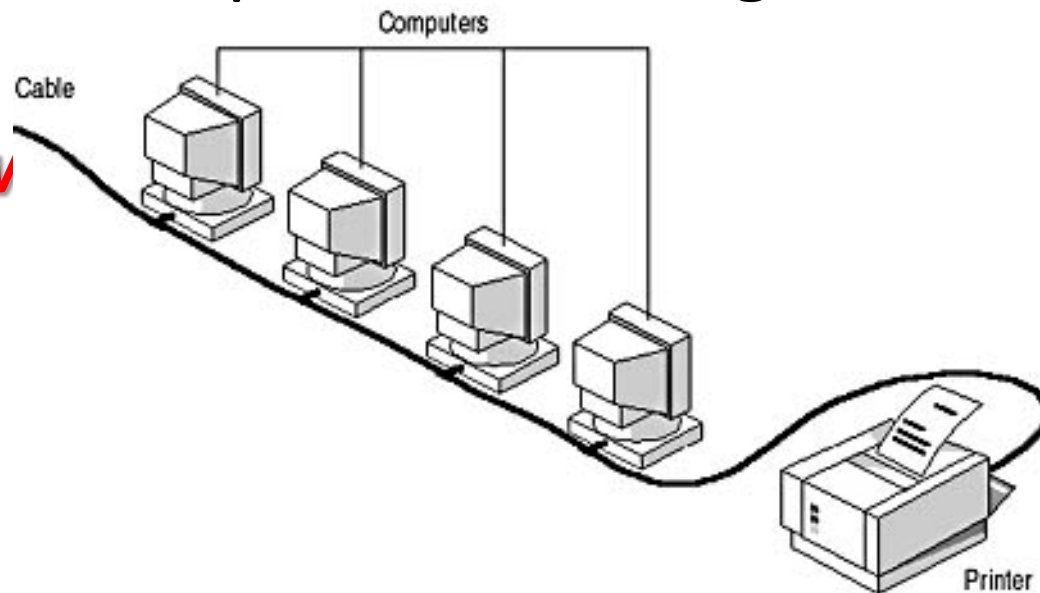
📢 This was, and still is, known as "working in a stand-alone environment."

📢 Copying files onto floppy disks and giving them to others to copy onto their computers was sometimes referred to as the "**sneakernet**." This is a very form of computer networking is one that many people have used and people



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- connecting together of computers and other devices is called a **network**, and the concept of connected computers sharing resources is called **netv**



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- Two computers are said to be interconnected if they are able to exchange information
- Components of a computer network

Hardware:

- *Computer*
- *Network card*
- *Routers*
- *Modem ...*

Media:

- *Cable*
- *Wire*
- *Microwave ...*

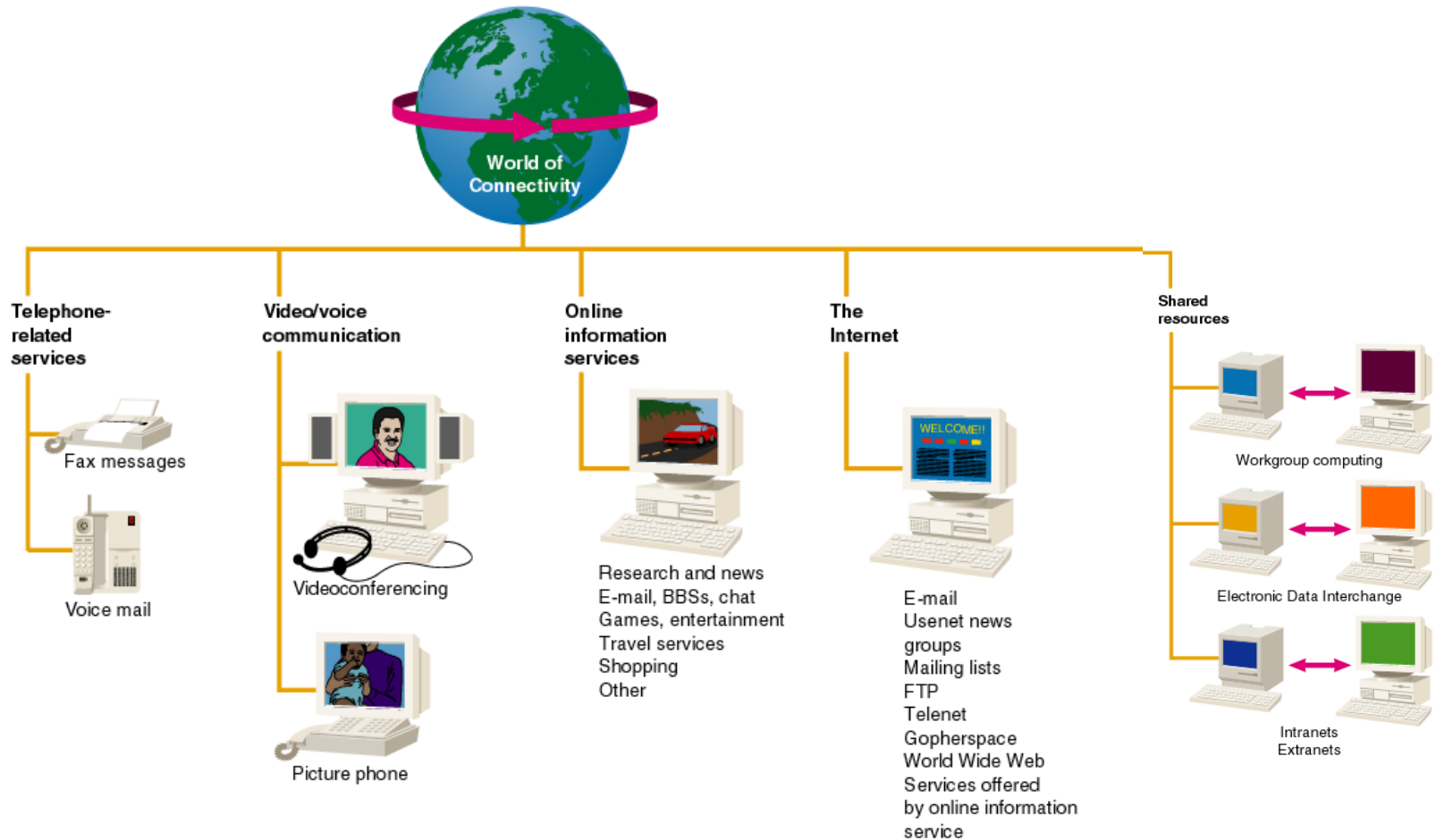
Software:

- *Network OS*
- *Utilities ...*

Network Design:

- *Logical layout*
- *Physical layout ...*

Uses of Computer Networks



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1. Resource sharing

- 📢 the goal of network is to make all programs, equipment, and especially data available to anyone on the network without regard to the physical location of the resource and the user.
- 📢 An obvious and widespread example is having a group of office workers share a common **printer**.
- 📢 None of the individuals really needs a private printer, and a high-volume networked printer is often cheaper, faster, and easier to maintain than a large collection of individual printers.
- 📢 **Information sharing** is more important than physical resource sharing

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2. Means of communication

- E-mail
- Videoconferencing
- Chatting
- E-commerce
- Game
-

3. Centralizing administration and support

- Database
- Banks
-

Classification of Computer Networks

- Computer networks are classified based on various factors. They include:
 - Administration
 - From an administrator's point of view, a network can be private network which belongs a single autonomous system and cannot be accessed outside its physical or logical domain.
A network can be public, which is accessed by all.
 - Geographical span
 - Connection Links
 - Inter-connectivity (Topology)
 - Architecture

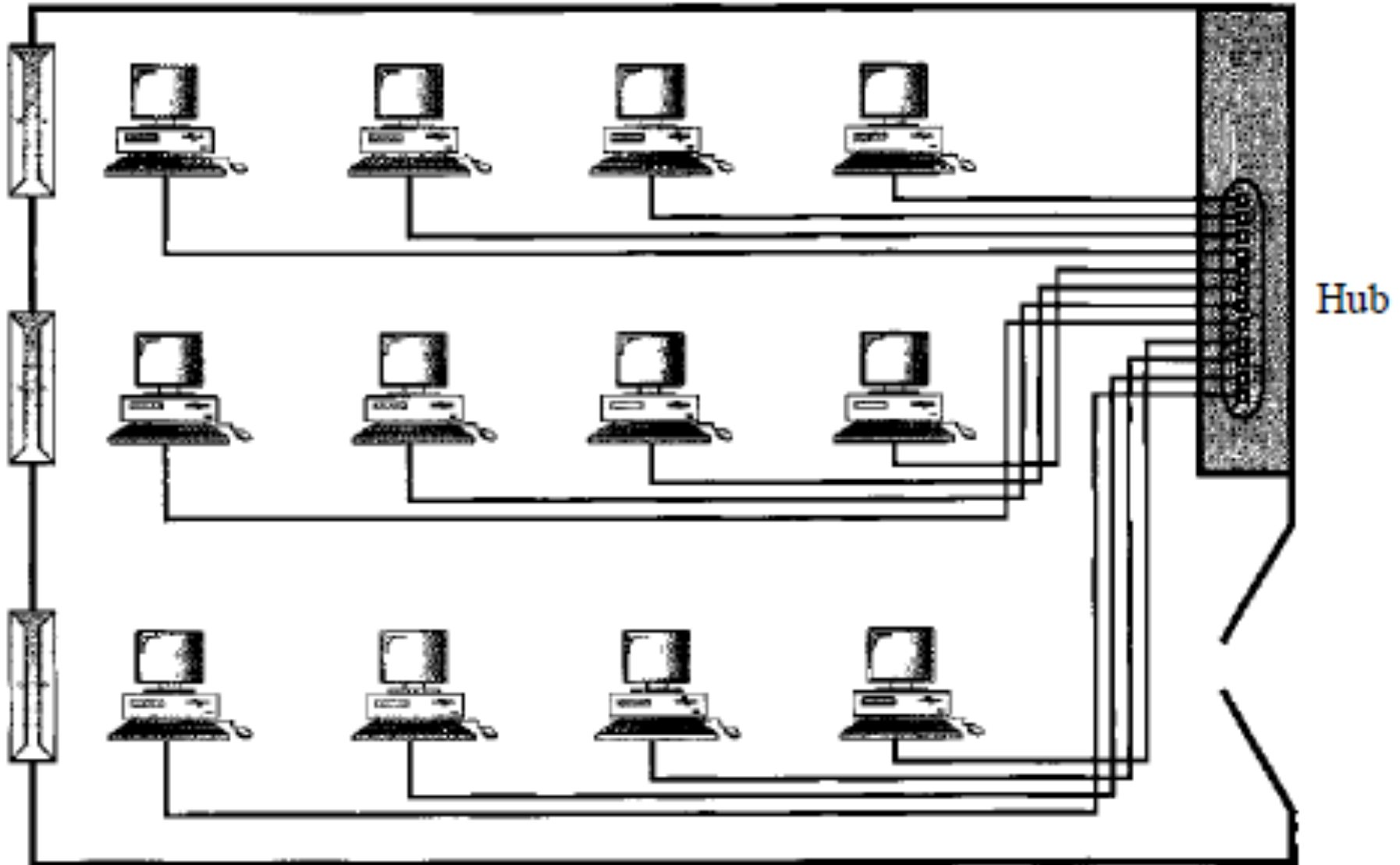
Network Types - Geographical Span

- 📌 Geographically a network can be seen in one of the following categories:
 - 📌 It may be spanned across your table, among Bluetooth enabled devices, Ranging not more than few meters.
 - 📌 It may be spanned across a whole building, including intermediate devices to connect all floors.
 - 📌 It may be spanned across a whole city.
 - 📌 It may be spanned across multiple cities or provinces.
 - 📌 It may be one network covering whole world.
- Generally there are three broad categories of networks based on its **size and geographical coverage**
 1. Local Area Network (LAN)
 2. Metropolitan Area Network (MAN)
 3. Wide Area Network (WAN)

LAN

- 📌 A **local area network (LAN)** is the basic building block of any computer network.
- 📌 A LAN can range from simple (two computers connected by a cable) to complex (hundreds of connected computers and peripherals throughout a major corporation).
- 📌 The distinguishing feature of a LAN is that it is confined to a **limited geographic area**.
- 📌 A local area network (LAN) is usually privately owned and links the devices in a single office, building, or campus.
- 📌 Depending on the needs of an organization and the type of technology used, a LAN can be as simple as two PCs and a printer in someone's home office; or it can extend throughout a company and include audio and video peripherals.
- 📌 Currently, LAN size is limited to a few **kilometers**

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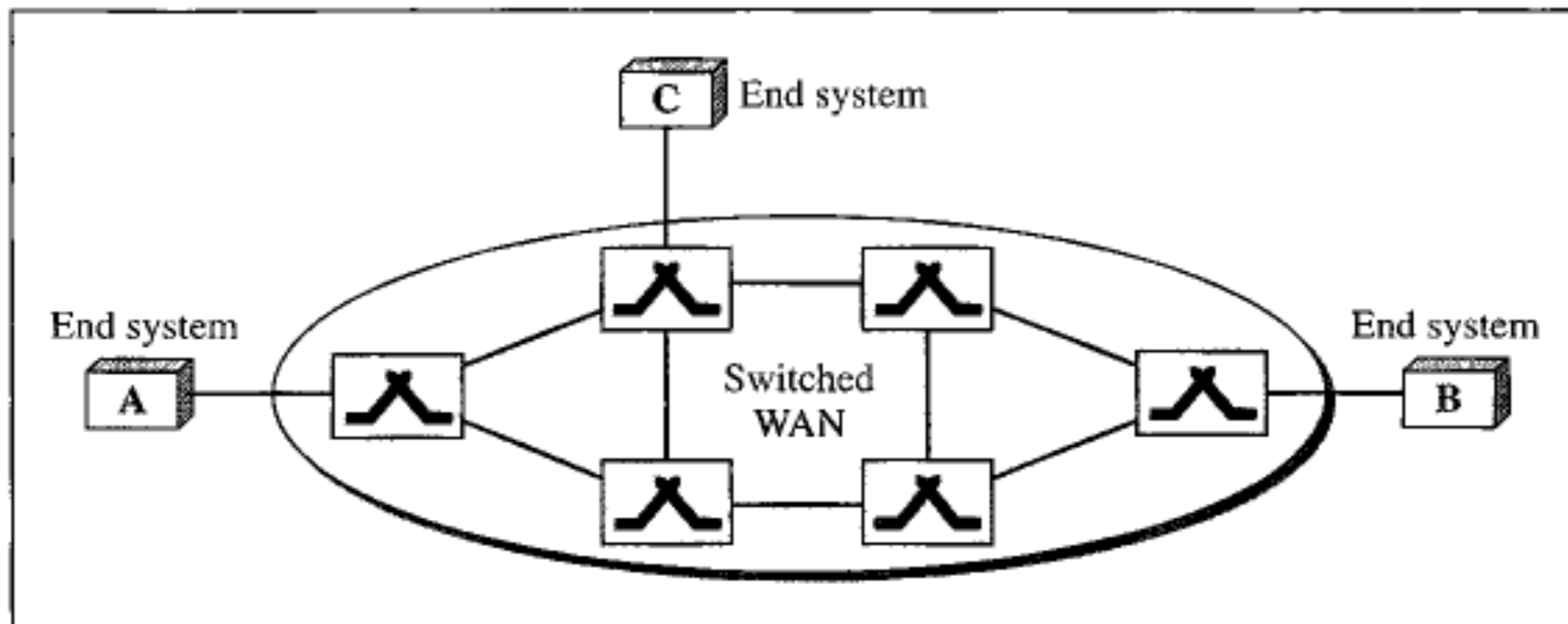
- ✧ LANs are designed to allow resources to be shared between personal computers or workstations. The resources to be shared can include hardware (e.g., a printer), software (e.g., an application program), or data.
- ✧ In addition to size, LANs are distinguished from other types of networks by their **transmission media** and **topology**.
- ✧ In general, a given LAN will use only **one type of transmission medium**. The most common LAN topologies are bus, ring, and star.
- ✧ Early LANs had data rates in the 4 to 16 megabits per second (Mbps) range. Today, however, speeds are normally 100 or 1000 Mbps.
- ✧ Wireless LANs are the newest evolution in LAN technology.

WAN

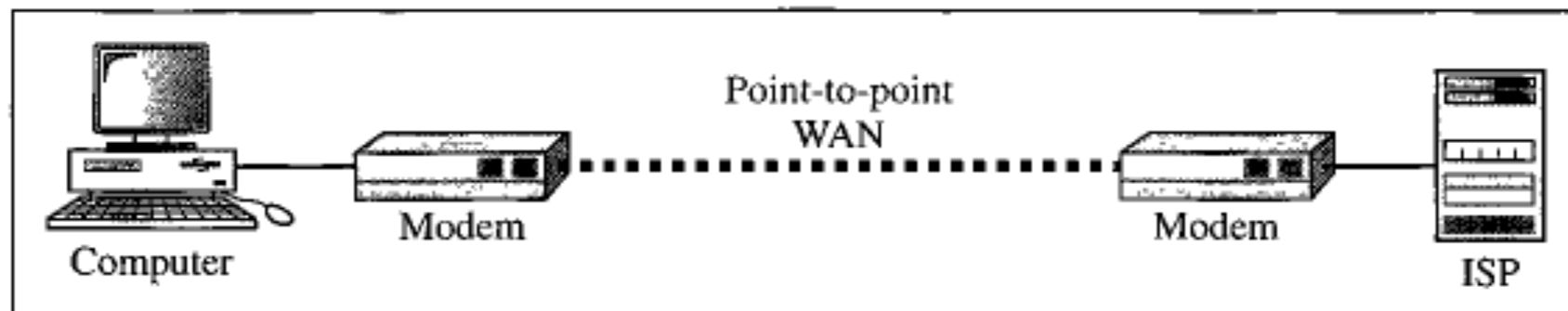
- A **wide area network (WAN)** provides long-distance transmission of data, image, audio, and video information over large geographic areas that may comprise a country, a continent, or even the whole world.
- A WAN can be as complex as the **backbones** that connect the Internet or as simple as a **dial-up line** that connects a home computer to the Internet. We normally refer to the first as a **switched WAN** and to the second as a **point-to-point WAN**

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- The **switched WAN** connects the end systems, which usually comprise a router (internetworking connecting device) that connects to another LAN or WAN.
- The **point-to-point WAN** is normally a line leased from a telephone or cable TV provider that connects a home computer or a small LAN to an Internet service provider (ISP). This type of WAN is often used to provide Internet access.



a. Switched WAN



b. Point-to-point WAN

MAN

- A metropolitan area network (MAN) is a network with a size between a LAN and a WAN.
- It normally covers the area inside a **town or a city**.
- It is designed for customers who need a **high-speed connectivity**, normally to the Internet, and have endpoints spread over a city or part of city.
- A good example of a MAN is the part of the telephone company network that can provide a high-speed **DSL line** to the customer (we will discuss DSL in later chapters) .
- Another example is the **cable TV** network that originally was designed for cable TV, but today can also be used for high-speed data connection to the Internet.

Summary of Networks Types

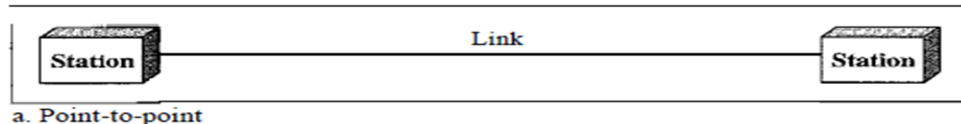
- As a general we can distinguish the difference of network type, based on the following key points
 - The geographical span they cover
 - Type of media they used
 - Data transmission rate
 - Type of service they provide
 - Cost

Network Type Based on Connection Links

- A network is two or more devices connected through **links**.
- A link is a communications pathway that transfers data from one device to another.
- For visualization purposes, it is simplest to imagine any link as a line drawn between two points.
- For communication to occur, two devices must be connected in some way to the same link at the same time.
- There are two possible types of connections: **point-to-point** and **multipoint**.

1. Point-to-Point

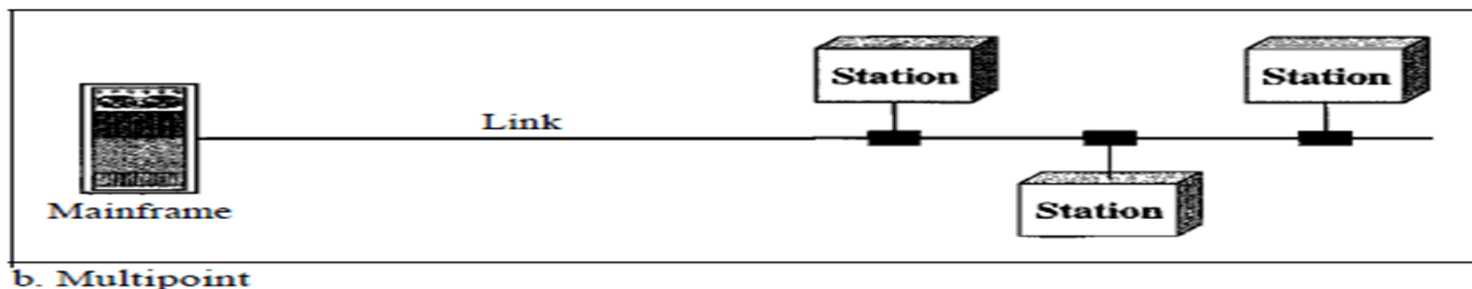
- A point-to-point connection provides a **dedicated link** between two devices.
- The entire capacity of the link is reserved for transmission between those two devices.
- Most point-to-point connections use an *actual length of wire or cable* to connect the two ends, but other options, such as *microwave or satellite links*, are also possible.



- When you change television channels by **infrared remote control**, you are establishing a point-to-point connection between the remote control and the television's control system.

2. Multipoint

- A multipoint (also called multidrop) connection is one in which more than two specific devices **share a single link**.
- In a multipoint environment, the capacity of the channel is shared, either **spatially** or **temporally**.
- If several devices can use the link simultaneously, it is a **spatially shared** connection.
- If users must take turns, it is a **timeshared connection**.



Network Topology (Inter - Connectivity)

- *Components of a network can be connected to each other differently in some fashion.*
 - *By connectedness we mean either logically, physically, or both ways.*
 - *Every single device can be connected to every other device on network, making the network mesh.*
 - *All devices can be connected to a single medium but geographically disconnected, created bus-like structure.*

Network Topology (Inter - Connectivity)

- *Each device is connected to its left and right peers only, creating linear structure*
- *All devices connected together with a single device, creating star-like structure.*
- *All devices connected arbitrarily using all previous ways to connect each other, resulting in a hybrid structure.*

Cont . . .

- A Network Topology is the arrangement with which computer systems or network devices are connected to each other.
- The term **topology** *refers to the way in which a network is laid out physically. (instead **Physical layout, Design, Diagram, Map** are also used)*
- Topologies may define **both physical and logical aspect** of the network.

Cont . . .

- Both logical and physical topologies could be same or different in a same network.
- The topology of a network is the **geometric representation** of the *relationship of all the links and linking devices* (usually called **nodes**) to one another.
- There are four basic topologies possible: **mesh, star, bus, and ring**

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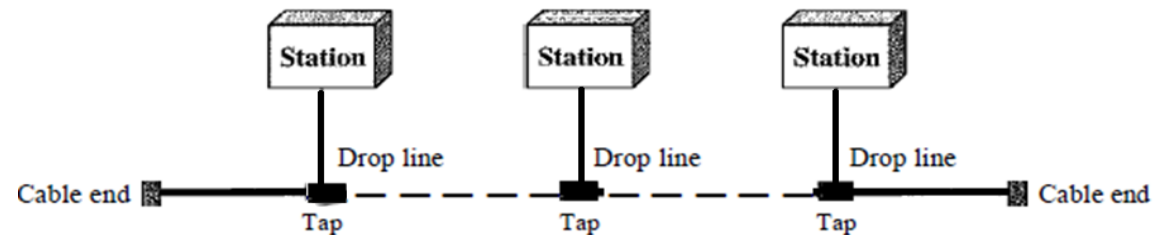
- The type of topology you use affects the type and capabilities. In general the choice of one topology over another will have an impact on the:
 - Type of equipment that the network needs
 - Capabilities of the network
 - Growth of the network
 - Way the network is managed

Cont . . .

- Developing a sense of how to use the different topologies is a key to understanding the capabilities of the different types of networks.
- Before computers can share resources or perform other communication tasks they must be connected. Most networks use cable to connect one computer to another. However, it is not as simple as just plugging a computer into a cable connecting to other computers

A. Bus Topology

- A bus topology, is **multipoint connection**.
- One long cable acts as a **backbone** to link all the devices in a network
- Nodes are connected to the bus cable by **drop lines** and **taps**.
- A **drop line** is a connection running between the device and the main cable.



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- A **tap** is a connector that either **splices into** the main cable or punctures the sheathing of a cable to create a contact with the **metallic core**.
- As a signal travels along the backbone, some of its energy is transformed into heat.
- Therefore, it becomes weaker and weaker as it travels farther and farther.
- For this reason there is a limit on the number of taps a bus can support and on the distance between those taps.

Advantages of Bus Topology.

- **Advantages** of a bus topology include **ease of installation**.
- Backbone cable can be laid along the **most efficient path**, then connected to the nodes by drop lines of various lengths.
- In this way, a bus **uses less cabling** than mesh or star topologies.
- *In a star, for example, four network devices in the same room require four lengths of cable reaching all the way to the hub.*
- *In a bus, this redundancy is eliminated. Only the backbone cable stretches through the entire facility.*
- Each **drop line has to reach only as far as the nearest point** on the backbone.

Disadvantages of Bus Topology.

1. **Difficult reconnection** and **fault isolation**. A bus is usually designed to be optimally efficient at installation.
 - It can therefore be difficult to add new devices.
2. **Signal reflection(bouns)** at the taps can cause degradation in quality. This degradation can be controlled by limiting the number and spacing of devices connected to a given length of cable.
 - Adding new devices may therefore require modification or replacement of the backbone.

Disadvantages of Bus Topology.

- In addition, a fault or break in the bus cable **stops all transmission**, even between devices on the same side of the problem.
- The damaged area reflects signals back in the direction of origin, creating noise in both directions.
- Bus topology was one of **the first topologies** used in the design of early local area networks. **Ethernet LANs can use a bus topology, but they are less popular now.**

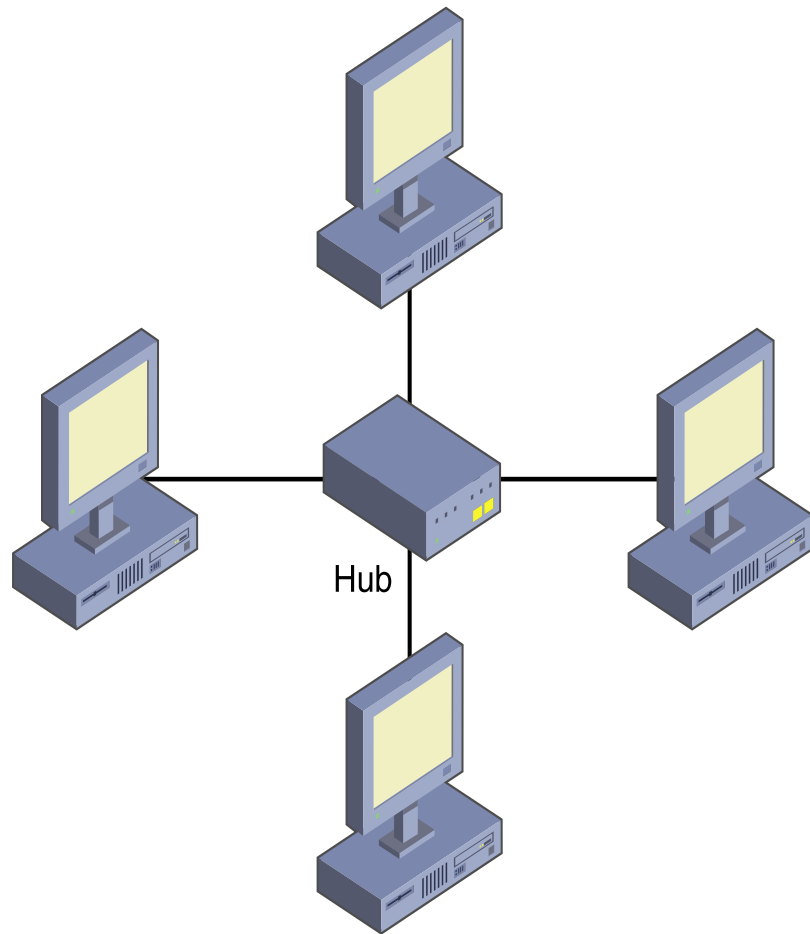
B. Star Topology

- In a **star topology**, each device has a dedicated **point-to-point** link only to a central controller, usually called a hub/switch.
- The term ***dedicated*** means that the link carries traffic only between the two devices it connects.
- The devices are not directly linked to one another.
- Unlike a mesh topology, a star topology does not allow direct traffic between devices.
- **The controller acts as an exchange:** If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device

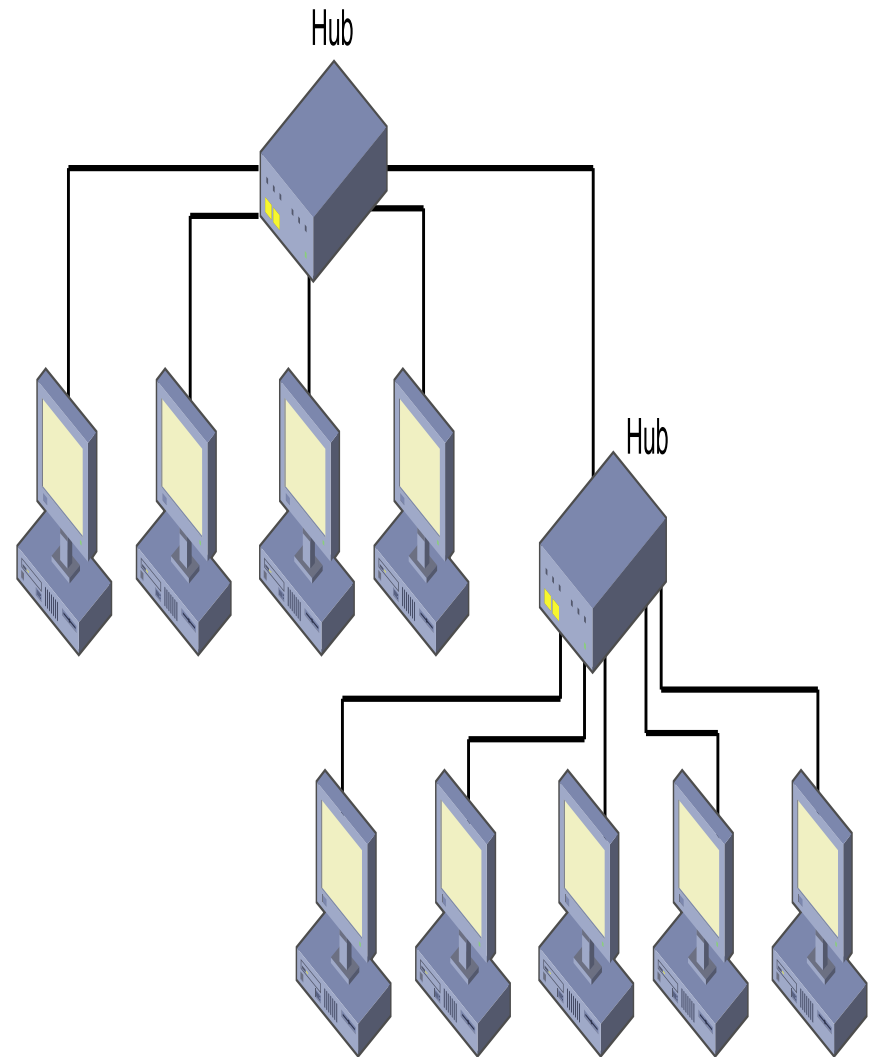
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- A star topology is less expensive than a mesh topology.
- In a star, each device needs only one link and one I/O port to connect it to any number of others.
- This factor also makes it **easy to install and reconfigure**.
- Far **less cabling** needs to be housed, and additions, moves, and deletions involve **only one connection**: between that device and the hub.
- Other advantages include **robustness**. If one link fails, only that link is affected. All other links remain active. This factor also lends itself to **easy fault identification** and **fault isolation**. As long as the hub is working, it can be used to monitor link problems and bypass defective links.

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A SIMPLE STAR TOPOLOGY



A HIERARCHICAL STAR TOPOLOGY

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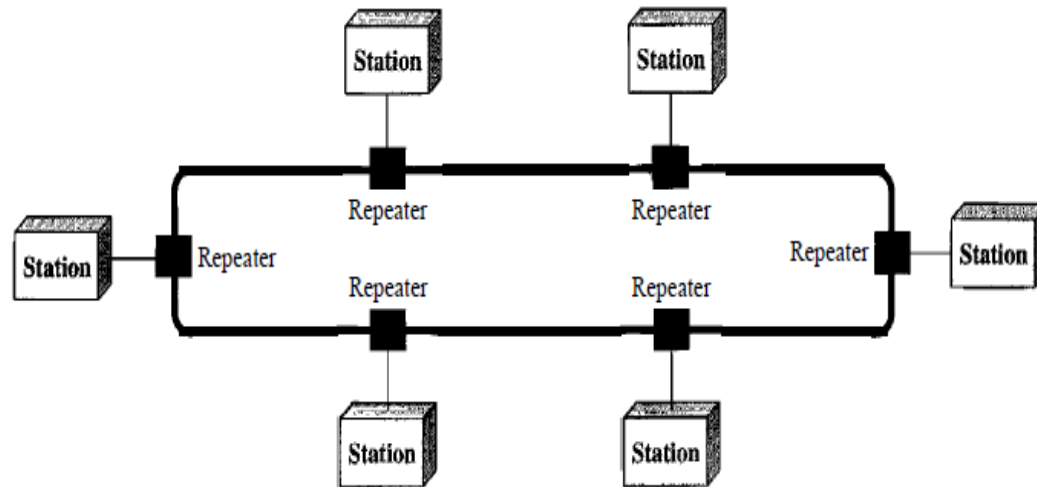
- One big disadvantage of a star topology is the **dependency of the whole topology on one single point**, the hub.
- If the hub goes down, the whole system is dead.
- Although a star requires far less cable than a mesh, each node must be linked to a central hub.
- For this reason, **often more cabling is required** in a star than in some other topologies (such as ring or bus).
- The star topology is used in local-area networks (LANs).
- High-speed LANs often use a star topology with a central hub.

C. Ring Topology

- In a **ring topology**, each device has a **dedicated point-to-point** connection with only the two devices on either side of it.
- A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- Each device in the ring incorporates a **repeater**.
- When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along

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- A ring is relatively easy to install and reconfigure.
- Each device is linked to only its immediate neighbors (either physically or logically).
- To add or delete a device requires changing only two connections.



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- The only constraints are media and traffic considerations (**maximum ring length** and **number of devices**).
- In addition, fault isolation is simplified.
- Generally in a ring, a signal is circulating at all times.
- In a simple ring, a **break in the ring** (such as a disabled station) can disable the entire network.
- This weakness can be solved by using a **dual ring** or a switch capable of closing off the break.
- Ring topology was prevalent when IBM introduced its local-area network **Token Ring**.
- Today, the need for higher-speed LANs has made this topology less popular.

D. Mesh topology

- In a **mesh topology**, every device has a dedicated point-to-point link to every other device.
- To find the number of physical links in a fully connected mesh network with n nodes, we first consider that each node must be connected to every other node.
- **Node 1** must be connected to $n - 1$ nodes, **node 2** must be connected to $n - 1$ nodes, and finally **node n** must be connected to $n - 1$ nodes. We need $n(n - 1)$ physical links.

D. Mesh topology

- However, if each physical link allows communication in both directions (duplex mode), we can divide the number of links by 2. In other words, we can say that in a mesh topology, we need $n(n-1)/2$ duplex-mode links.
- To accommodate that many links, every device on the network must have $n - 1$ input/output (I/O) ports to be connected to the other $n - 1$ stations.

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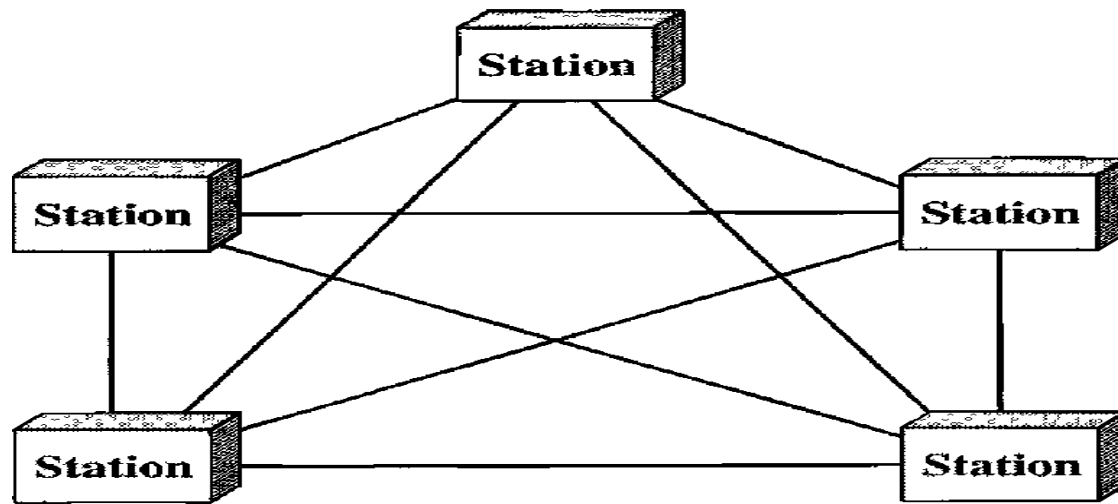
- A mesh offers several advantages over other network topologies
- First, the use of dedicated links guarantees that each connection can carry its own data load, thus **eliminating the traffic problems** that can occur when links must be shared by multiple devices.
- Second, a mesh topology is **robust**. If one link becomes unusable, it does not incapacitate the entire system.
- Third, there is the advantage of **privacy** or **security**. When every message travels along a dedicated line, only the intended recipient sees it. Physical boundaries prevent other users from gaining access to messages.
- Finally, point-to-point links make **fault identification and fault isolation easy**.
- Traffic can be routed to avoid links with suspected problems. This facility enables the network manager to discover the precise location of the fault and aids in finding its cause and solution.

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- The main disadvantages of a mesh are related to the **amount of cabling and the number of I/O ports required**.
- First, because every device must be connected to every other device, **installation and reconnection are difficult**.
- Second, the **sheer bulk of the wiring** can be greater than the available space (in walls, ceilings, or floors) can accommodate.
- Finally, the hardware required to connect each link (I/O ports and cable) can be **prohibitively expensive**.
- For these reasons a mesh topology is usually implemented in a **limited fashion**, for example, as a backbone connecting the main computers of a hybrid network that can include several other topologies.

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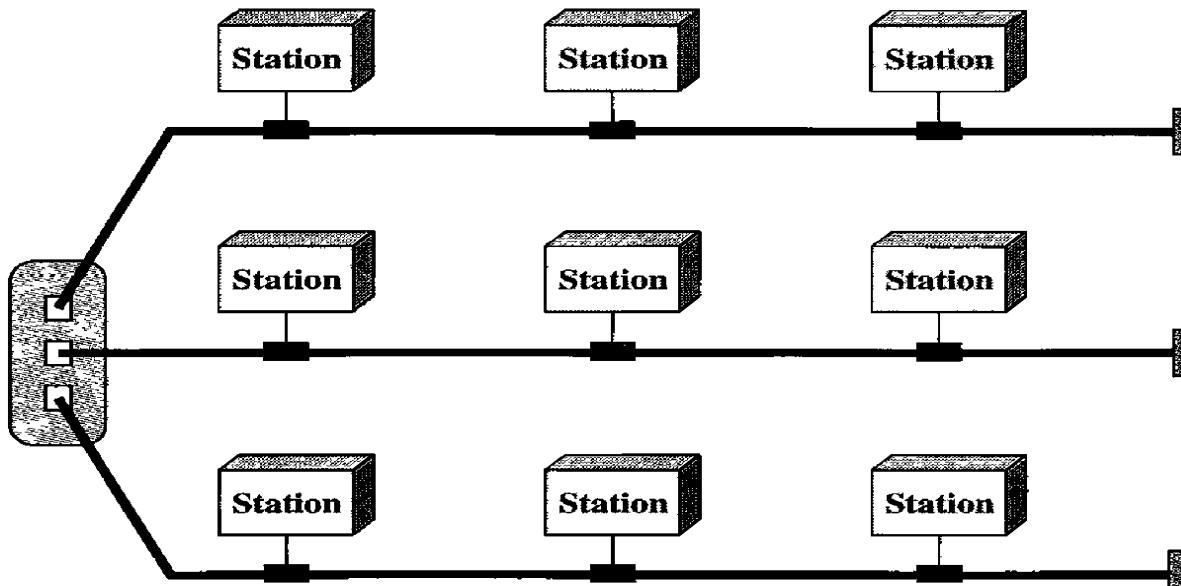
- One practical example of a mesh topology is the connection of **telephone regional offices** in which each regional office needs to be connected to every other regional office



Mesh topology

E. Hybrid Topology

- A network can be hybrid. For example, we can have a main star topology with each branch connecting several stations in a bus topology (**star-bus topology**)



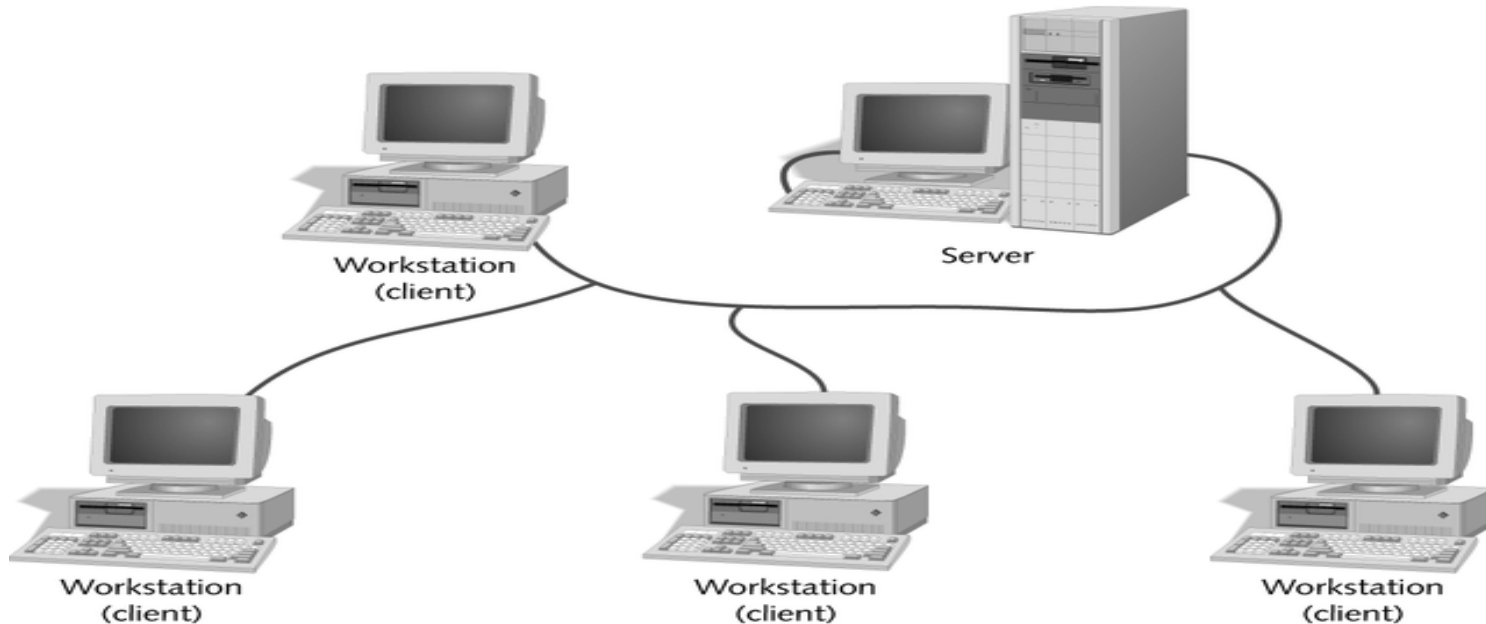
Network Models (Architecture) by Capability

- Computer networks can be discriminated into various types such as Client-Server, peer-to-peer or hybrid, depending upon its architecture.
 - ✓ There can be one or more systems acting as Server. Other being Client, requests the Server to serve requests. Server takes and processes request on behalf of Clients.
 - ✓ Two systems can be connected Point-to-Point, or in back-to-back fashion. They both reside at the same level and called peers.
- There can be hybrid network which involves network architecture of both the above types.
 1. Client/Server Model
 2. Peer-to- Peer Model

Network Models by Capability

1. Client/Server Model

- Microcomputer users, or *clients*, share services of a centralized computer called a *server*.



Client-server model

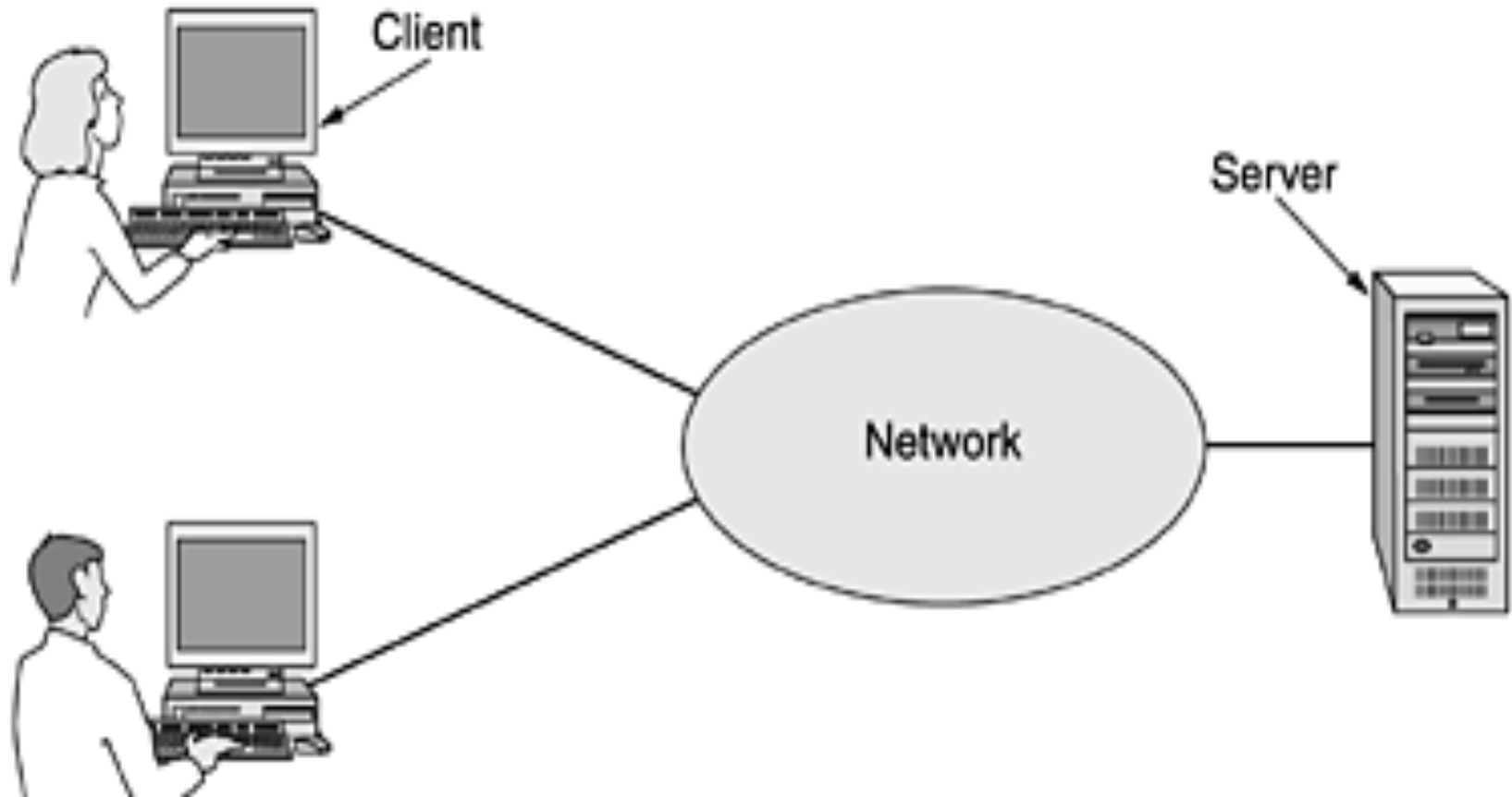


Fig. Two clients using a single server at a time

Client/Server

- A distinction exists between computers that make available network resources (servers) and those computers that use the resources (clients or workstations).

Pros:

- Very secure
 - Centralized servers easy to manage
 - Physically centralized
 - Secure OS
- Better performance
- Centralized backups
- Reliability
 - Simple job to do plus built in redundancy

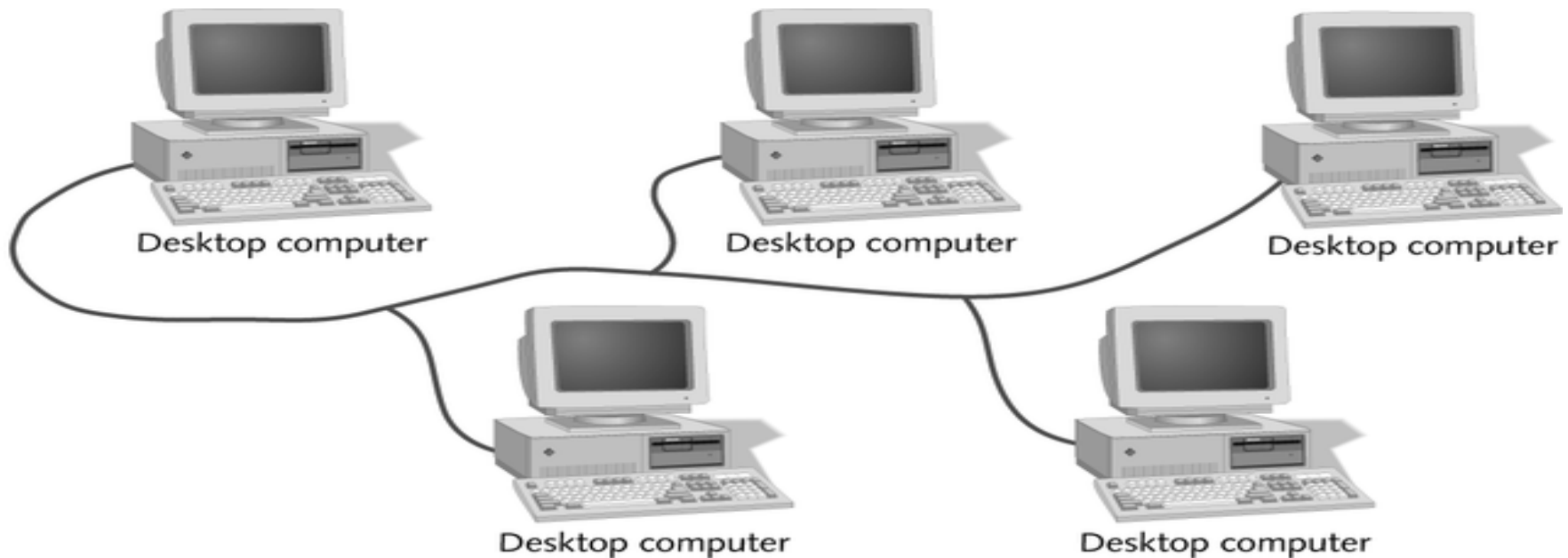
Cons:

- Require professional administration
- More hardware intensive

Network Models by Capability

2. Peer-to-Peer Model

- Computers share equally with one another without having to rely on a central server.



Peer-to-Peer (p2p)

- Computers on the network communicate with each others as equals and each computer is responsible for making its own resources available to other computers on the network.

Pros:

- Uses less expensive computer networks
- Easy to administer
- No NOS (network operating system) required
- More built-in redundancy
 - Shared resources – some machine will have what you need

Peer-to-Peer (p2p)

Cons:

- Individual user performance easily affected
- Not very secure
 - Tragedy of the commons – no guarantee others will administer their resources properly (almost guaranteed with over 10 machines)
- Hard to back up.

Client/Server vs. Peer-to-Peer: Advantages and Disadvantages

Client/Server Model

Advantages:

- Very secure OS.
- Better performance.
- Centralized servers, easy to manage.
- Centralized backups.
- High reliability.

Disadvantages:

- Expensive administration.
- More hardware intensive.

Peer-to-Peer Model

Advantages:

- Uses less expensive networks.
- Easy to administer.
- Contain both network operating system and application software.
- Ideal for small business and home users (up to 10 computers).

Disadvantages:

- Individual user performance easily affected.
- Not very secure.
- Hard to back up.

Summary of Networks Models

- We can distinguish network models/architecture difference based on the following key points.
 - » Their functions
 - » Security level
 - » Scalability
 - » Type of operating system they used
 - » Backup process

Network Criteria

- A network must be able to meet a certain number of criteria.
- The most important of these are :-
 - Performance,
 - Reliability
 - Security.

Network Criteria

■ Performance

- Can be measured in many ways, including transmit time and response time.
- ***Transmit time*** is the amount of time required for a message to travel from one device to another.
- ***Response time*** is the elapsed time between an inquiry and a response.
- Depends on a number of factors:
 - » the ***number of users***
 - » ***the type of transmission medium***
 - » the ***capabilities of the connected hardware***
 - » ***the efficiency of the software.***

Network Criteria

- **Reliability**

- In addition to accuracy of delivery, network reliability is measured by the *frequency of failure*, the *time* it takes a link to *recover* from a failure.

- **Security**

- Network security issues include :
 - protecting data from unauthorized access
 - protecting data from damage and development
 - implementing policies and procedures for recovery from breaches and data losses.

Standards

- ✈ Most networks are a combination of hardware and software from a variety of vendors.
- ✈ The ability to combine the products manufactured by different vendors is made possible by the existence *of industry standards*.
- ✈ ***Standards are guidelines*** that vendors **adhere /stick on** to voluntarily in order to make their products compatible with products from other vendors.
- ✈ In general those standards address:
 - Size.
 - Shape.
 - Material.
 - Function.
 - Speed. Etc.

Contd.

- ✉ More specifically, the standards define physical and operational characteristics of:
 - Operating systems (NOS)
 - Communication Software.
- ✉ For example, standards make it possible to buy a NIC manufactured by one vendor for a computer manufactured by another vendor with reasonable assurance that the card will:
 - Fit into the computer.
 - Work with the network cabling.
 - Translate signals from computer and send them out onto network.
 - Receive data from the network and deliver it to the computer.
- ✉ When a vendor subscribes to a set of standards, it means that the vendor is agreeing to make equipment that conforms to the specifications of the standard.

Layered communication model

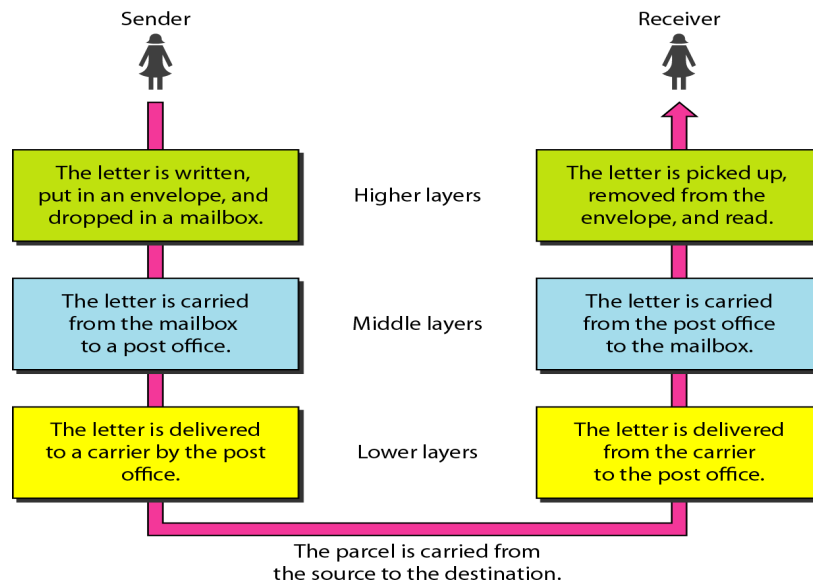
- ✚ Layers model is a theoretical approach that describe how data is going to be transmit over the network.
- ✚ There are several layer models available, among those following two models are the most popular:
 - ✚ OSI Reference model
 - ✚ TCP / IP Reference model
- ✚ A **communication architecture** is a strategy for connecting host computers and other communicating equipment by define all **necessary elements** for data communication between devices.

Layered Tasks

✈ We use the concept of layers in our daily life.

✈ As an example, let us consider **two friends** who communicate through **postal mail**.

✈ The process of sending a letter to a friend would be complex if there were **no services** available from the **post office**.



Contd.

- ✎ Layer architecture simplifies the **network design**.
- ✎ It is easy to **debug network applications** in a layered architecture network.
- ✎ The **network management** is easier due to the layered architecture.
- ✎ Layering communication functions adds **flexibility**, for example, we do **not need to modify** the entire host software to include more communication devices.
- ✎ Network layers follow a set of rules, called **protocol**.
- ✎ The protocol defines the **format of the data** being exchanged, and the **control and timing** for the **handshake** between layers.

Open Systems Interconnection (OSI)

- ✚ International standard organization (ISO) established a committee in 1977 to develop an architecture for computer communication.
- ✚ Open Systems Interconnection (OSI) reference model is the result of this effort.
- ✚ In 1984, the Open Systems Interconnection (OSI) reference model was approved as an international standard for communications architecture.
- ✚ Term “open” denotes the ability to connect any two systems which conform to the reference model and associated standards.
- ✚ The OSI model is now considered the **primary Architectural model** for inter-computer communications.

Contd.

- It is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture.
- **The Purpose:**
 - to show how to facilitate communication between different systems without requiring changes to the logic of the underlying hardware and software
- It is not a protocol;
 - it is a model for understanding and designing a network architecture that is flexible, robust, and interoperable.

Contd.

✉ The OSI model describes how information or data makes its way from an application programmes (such as spreadsheets) in computer, through a network medium (such as wire) to another application programme located in another computer or on another network.

✉ The model also helps to troubleshoot problems by providing a frame of reference that describes how components are supposed to function

Contd.

- ✚ The OSI reference model divides the **problem of moving information** between computers over a network medium into **SEVEN smaller** and more **manageable problems**.
- ✚ This separation into smaller more manageable functions is known as **layering**.
- ✚ The main objective of dividing into seven layers include creating a manageable layers by breaking down the complex network operations and allowing specialization of research and development on each component of the OSI layer

Contd.

- ✧ In *developing the model*, the designers distilled the process of **transmitting data to its most fundamental elements**.
- ✧ They identified which networking functions had related uses and collected those functions into discrete groups that became the layers.
- ✧ Each layer defines a family of functions distinct from those of the other layers.
- ✧ By defining and localizing functionality in this fashion, the designers created an architecture that is both **comprehensive and flexible**.
- ✧ Most importantly, the OSI model allows complete **interoperability** between otherwise incompatible systems.

Contd.

Purpose of OSI model

- ✧ OSI model presents a structured set of seven layers interconnecting as a stack.
- ✧ Each layer in this model was developed to maintain the ability of connecting the devices, computers and network segments.
- ✧ OSI model was created with following purposes:-
 - ✧ *To create a common platform for software developers and hardware manufactures that encourage the creation of networking products that can communicate with each other over the network.*
 - ✧ *To help network administrators by dividing large data exchange process in smaller segments. Smaller segments are easier to understand, manage and troubleshoot. With layer approach they only have to troubleshoot the devices which are working in faulty layer.*

Contd.

- ✧ **In general, the Benefits of the OSI Reference model is given below**
 - ✧ Big picture of communication over network is understandable through this OSI model.
 - ✧ OSI model distinguishes well between the services, interfaces and protocols.
 - ✧ Protocols of OSI model are very well hidden.
 - ✧ We see how hardware and software work together.
 - ✧ **Reduces complexity:-** Dividing network communication process in smaller component makes software development, design, and troubleshooting easier.
 - ✧ **Standardizes interfaces:-** It describes what functions occur at each layer of the model that encourages industry standardization.
 - ✧ **Facilitates modular/modification engineering:-** It prevents changes in one layer from affecting other layers.
 - ✧ **Ensures interoperable technology:-** It allows different types of network hardware and software to communicate. And can be used to compare basic functional relationships on different networks

Contd.

✧ In general, the **Benefits of the OSI Reference model** is given below

✧ **Accelerates evolution**:- Standardization of network components allows multiple-vendor development.

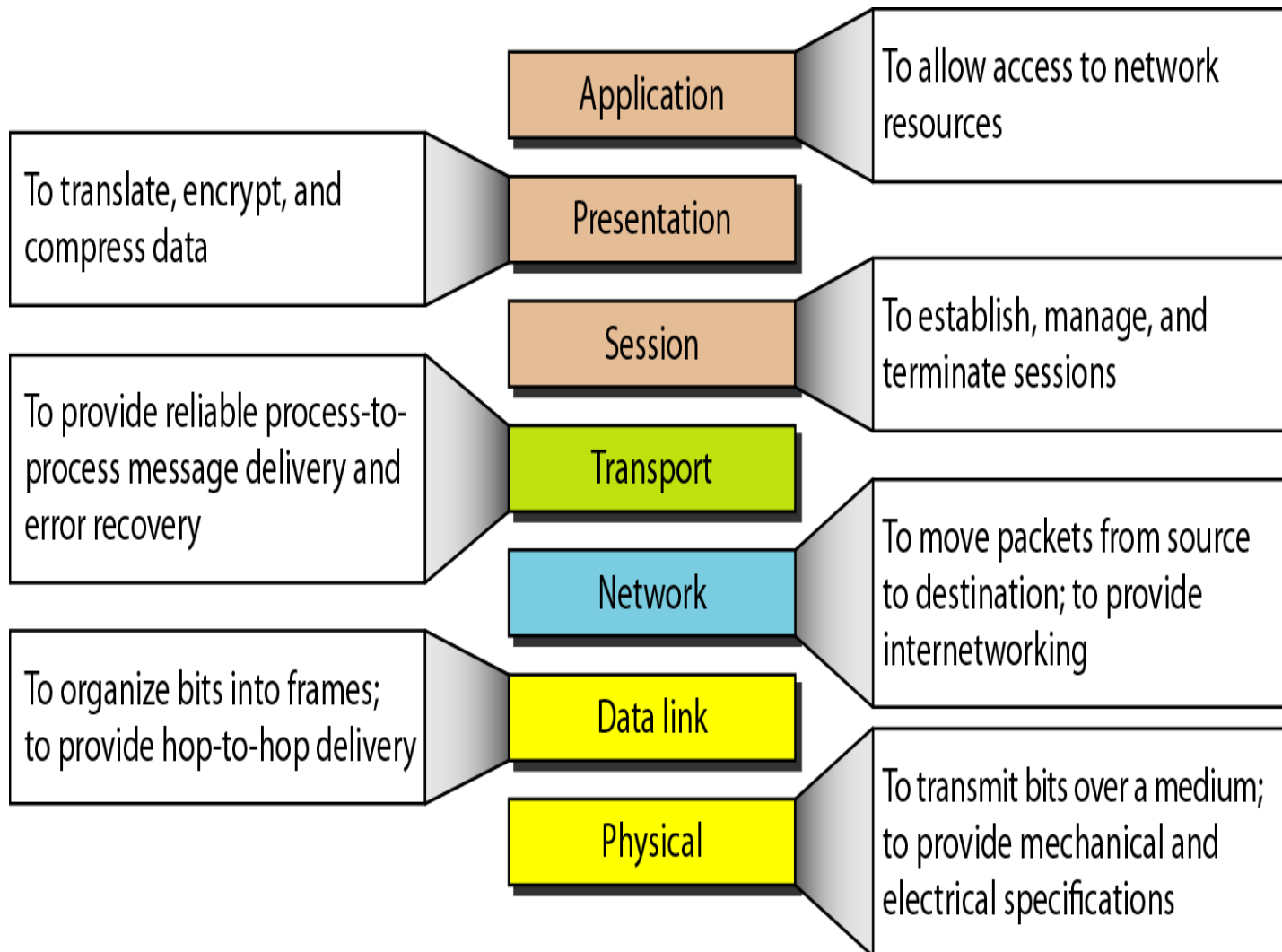
✧ **Quick troubleshooting**: Enable network administrators to troubleshoot issues more quickly and effectually by looking in layer that is causing issue rather than finding it entire network.

✧ **Simplifies teaching and learning**:- We can understand new technologies as they are developed.

■ **Demerits of OSI reference model:**

- Model was devised before the invention of protocols.
- Fitting of protocols is tedious task.
- It is just used as a reference model.

The Seven Layers



OSI Model

Application Layer

- enables the user(human or software) to access the network.

Presentation Layer

- concerned with the syntax and semantics of the information exchanged between two systems.

Session Layer

- The session layer is responsible for dialog control and synchronization.

Transport Layer

- responsible for process-to-process delivery of the entire message.

Network Layer

- responsible for the source-to-destination delivery of a packet, possibly across multiple networks (links).

Data Link Layer

- the delivery of the packet between two systems on the same network (links)

Physical Layer

- It deals with the mechanical and electrical specifications of the interface and transmission medium.

OSI Model

Application Layer

- it provides user interfaces and support for services such as email, ftp, shared database management, Telenet

Presentation Layer

- Translation ,Encryption , Compression

Session Layer

- Dialog control ,Synchronization

Transport Layer

- Service-point addressing ,Segmentation and reassembly, Connection control, Flow control ,Error control:

Network Layer

- Logical addressing ,Routing.

Data Link Layer

- Framing, Physical addressing, Flow control, Error control ,Access control

Physical Layer

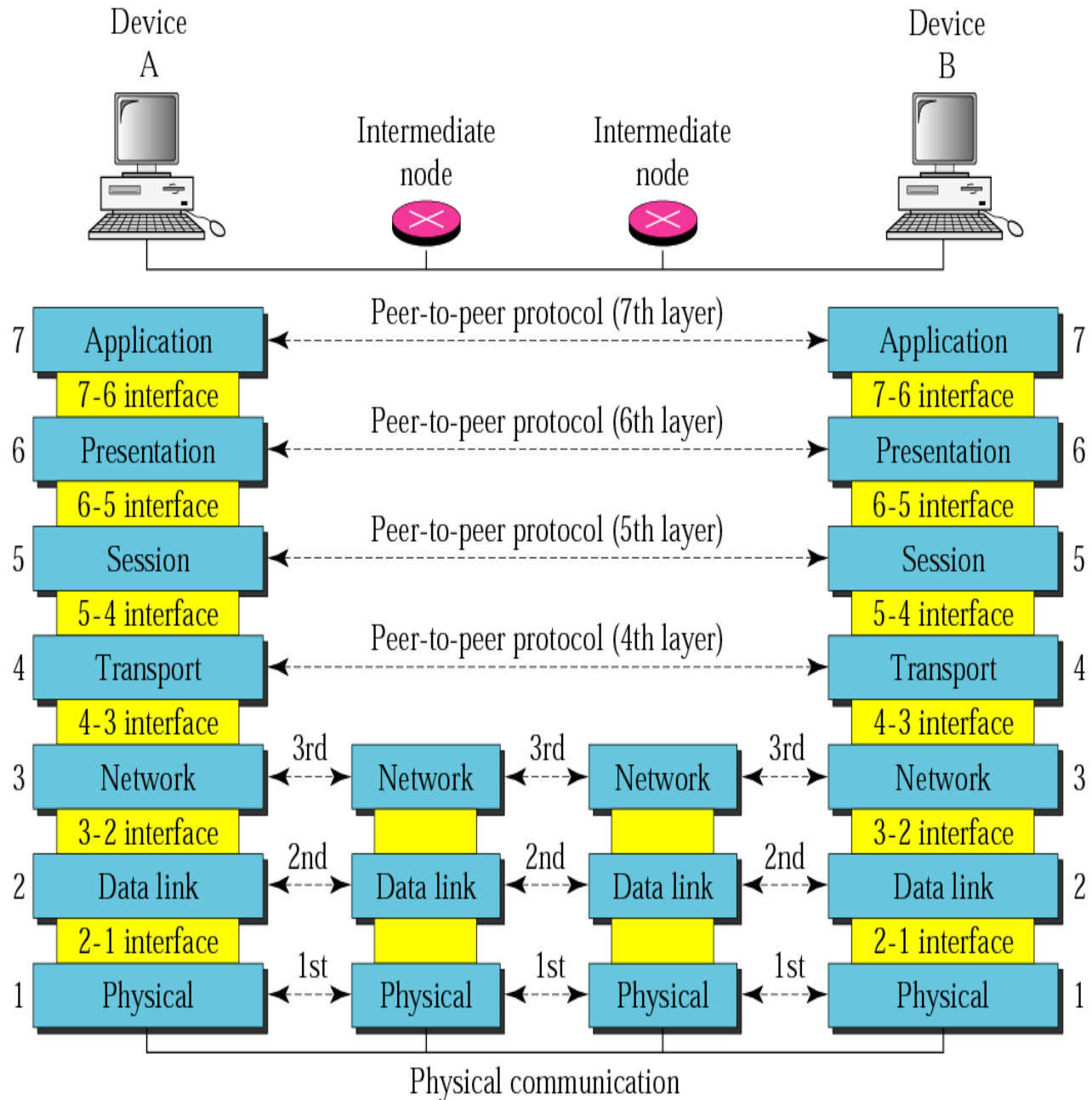
- Physical characteristics of interfaces and medium, Representation of bits ,Data rate ,Synchronization of bits ,Line configuration ,Physical topology, Transmission mode

Contd.

✉ The figure shows the layers involved when a message is sent from device A to device B.

✉ As the message travels from A to B, it may pass through many intermediate nodes.

✉ These intermediate nodes usually involve only the first three bottom layers of the OSI model.



Contd.

- Layer 3, for example, uses the services provided by layer 2 and provides services for layer 4. Between machines, layer x on one machine communicates with layer x on another machine.
- This communication is governed by an agreed-upon series of rules and conventions called protocols.
- The processes on each machine that communicate at a given layer are called peer-to-peer processes.

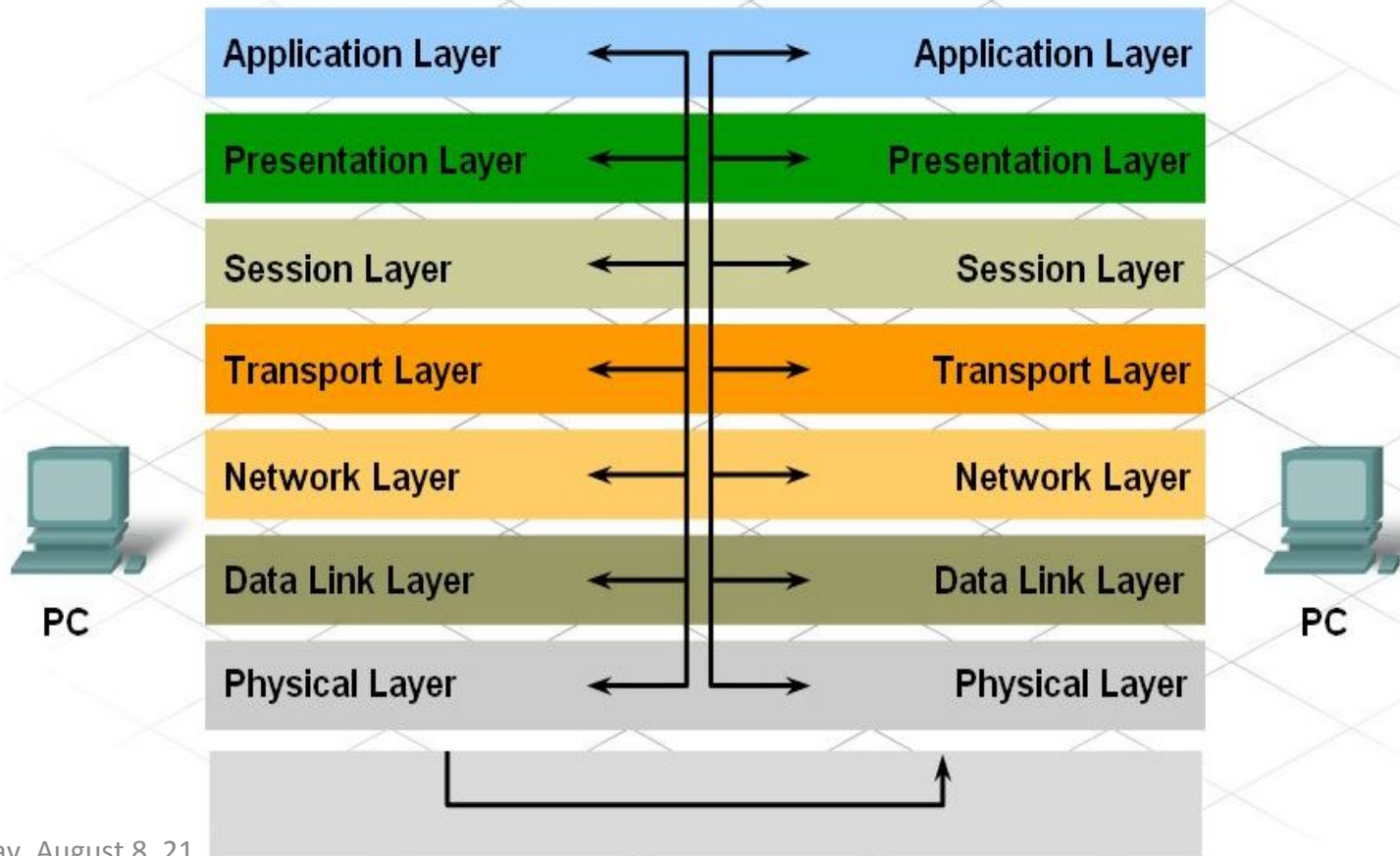
Organization of the Layers

- ✎ The seven layers can be thought of as belonging to **three subgroups**.
- ✎ Layers 1, 2, and 3 - **physical, data link, and network** - are the **network support layers**; they deal with the physical aspects of moving data from one device to another (such as **electrical specifications, physical connections, physical addressing, and transport timing and reliability**).
- ✎ Layers 5, 6, and 7- **session, presentation, and application** - can be thought of as the **user support layers**; they allow interoperability among unrelated software systems.
- ✎ Layer 4, the **transport layer**, **links** the two subgroups and **ensures that what the lower layers have transmitted is in a form that the upper layers can use**.
- ✎ The upper OSI layers are almost always implemented in **software**; lower layers are a combination of **hardware and software**, except for the physical layer, which is **mostly hardware**.

Contd.

✉ A given layer within the OSI model communicates with three other layers:

1. the layer directly above it,
2. the layer directly below it and
3. its peer layer in other networked computer system.



Data Exchange Process

- In data exchange process, participating computers work in reverse mode. Layers on receiving computer perform the same task in reverse mode.
- The receiving device takes delivery of, handles, and translates the data from the sending device at a particular layer.
- For example on sending computer presentation layer compress the data, same presentation layer on receiving computer decompress the data.

Contd.

On sending computer

- Sending application access the application layer.
- Application provides data to the presentation layer.
- Presentation layer format data as per network requirement and forward it's to session layer.
- Session layer initiate connection and forward the data to the transport layer.
- Transport layer broke down the large data file in smaller segments and add a header with control information, which are bits designated to describe how to determine whether the data is complete, uncorrupted, in the correct sequence, and so forth.
- Segments are forwarded to network layer. Network layer add its header, with logical address and convert it in packet and forwards packet to data link layer
- Data link layer attach its header and footer to packet and convert it in frame.
- Frames are forwarded to the physical layers that convert them in signals.
These signals are loaded in media.

Contd.

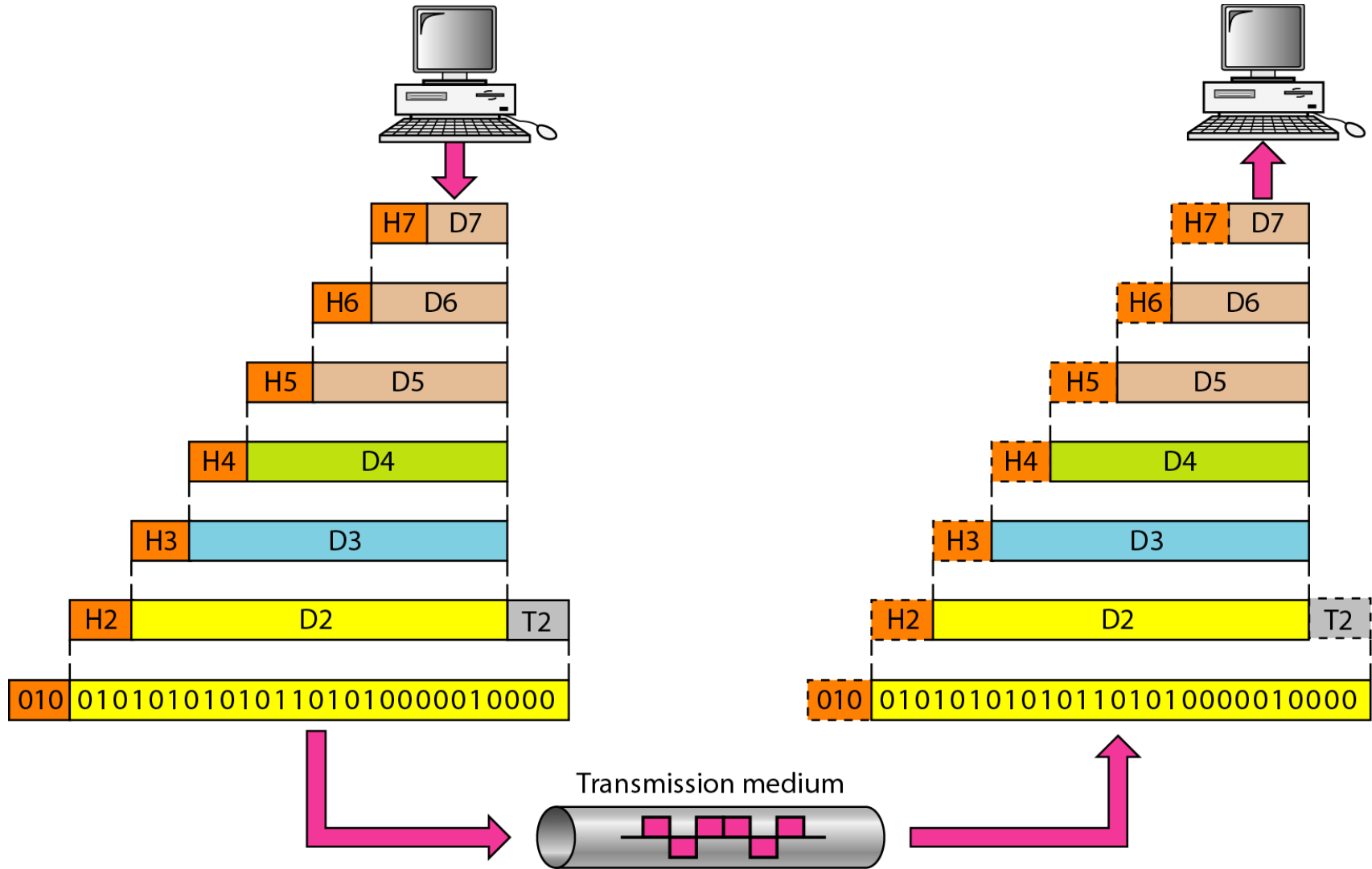
On receiving computer

- Physical layer receive signals from media and convert them in frames. Frames are forwarded to the data link layer.
- Data link layer check the frame. All tampered frame are dropped here. If frame is correct, data link layer strip down its header and footer from frame and hand over packet to network layer.
- Network layer check the packet with its own implementations. If it's found everything fine with packet, it strips down its header from packet and hand over segment to transport layer.
- Transport layer again do the same job. It verifies the segments with its own protocol rules. Only the verified segments are processed. Transport layer remove its header from verified segments and reassemble the segments in data. Data is handed over the session layer.
- Session layer keep track of open connection and forwarded the receiving data to presentation layer.
- Presentation form the data in such a way that application layer use it.
- Application layer on receiving computer find the appropriate application from the computer and open data within particular application.

Contd.

- ✎ The figure shown in the next slide, gives an overall view of the OSI layers, **D7** means the data unit at **layer 7**, **D6** means the data unit at **layer 6**, and so on. The process starts at **layer 7** (the application layer), then moves from layer to layer in descending, sequential order.
- ✎ At each layer, a **header**, or **possibly a trailer**, can be added to the data unit.
- ✎ Commonly, the trailer is added only at **Data-Link Layer (Layer 2)**.
- ✎ When the formatted data unit passes through the **physical layer**, it is changed into an **electromagnetic signal** & transported along a physical link.

Data exchange using the OSI model



The TCP/IP Reference Model

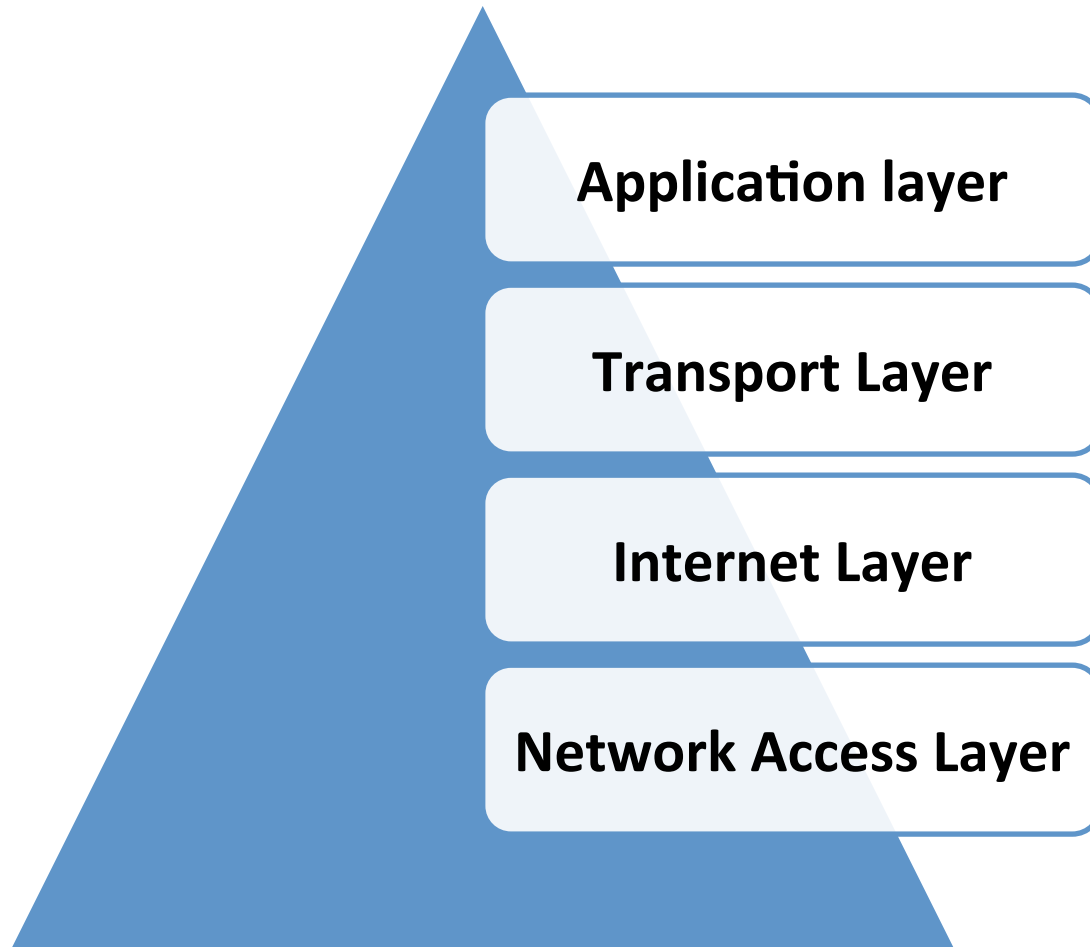
The TCP/IP Reference Model

- ✚ TCP/IP means Transmission Control Protocol and Internet Protocol.
- ✚ TCP/IP protocol model is another popular layer model that describes network standards.
- ✚ It was developed by Department of Defense's Project Research Agency (ARPA, later DARPA) as a part of a research project of network interconnection to connect remote machines.
- ✚ The features that stood out during the research, which led to making the TCP/IP reference model were:
 - *Support for a flexible architecture. Adding more machines to a network was easy.*
 - *The network was robust, and connections remained intact until the source and destination machines were functioning.*
- The overall idea was to allow one application on one computer to talk (to send data packets) another application running on different computer.

Cont . . .

- ✎ This model has same names of layers as OSI reference model has.
- ✎ Don't be confuse with same name, layers at both model have different functionality in each model.
- ✎ It is the network model used in the current Internet architecture as well.
- ✎ These protocols describe the movement of data between the source and destination or the internet.
- ✎ These protocols offer simple naming and addressing schemes.

TCP/IP Model



Overview of Functions of TCP/IP Layers

Application layer:

- TCP/IP model combine the functionality of application layer, presentation layer and session layer from OSI model in single application layer.
- In TCP/IP model application layer do all tasks those are performed by upper layers in OSI model.
- Application layer deals with high level protocols, including data presentation, compression and dialog control.
- The specifications of this model describe a lot of applications that were at the top of the protocol stack. Some of them were TELNET, FTP, SMTP, DNS
 - *TELNET: allows connecting to a remote machine and run applications on it.*
 - *FTP(File Transfer Protocol): allows File transfer amongst computer users connected over a network. It is reliable, simple and efficient.*
 - *SMTP(Simple Mail Transport Protocol): used to transport electronic mail between a source and destination, directed via a route.*
 - *DNS(Domain Name Server) resolves an IP address into a textual address for Hosts connected over a network.*

Cont . . .

- Transport layer:
 - TCP protocol is used for reliable data delivery.
 - Flow control and error correction methods are used for guaranteed data delivery.
 - It decides if data transmission should be on parallel path or single path.
 - Functions such as multiplexing, segmenting or splitting on the data
 - The applications can read and write to the transport layer.
 - Transport layer breaks the message (data) into small units so that they are handled more efficiently by the network layer.
 - Transport layer also arrange the packets to be sent, in sequence.

Cont . . .

■ Internet layer:

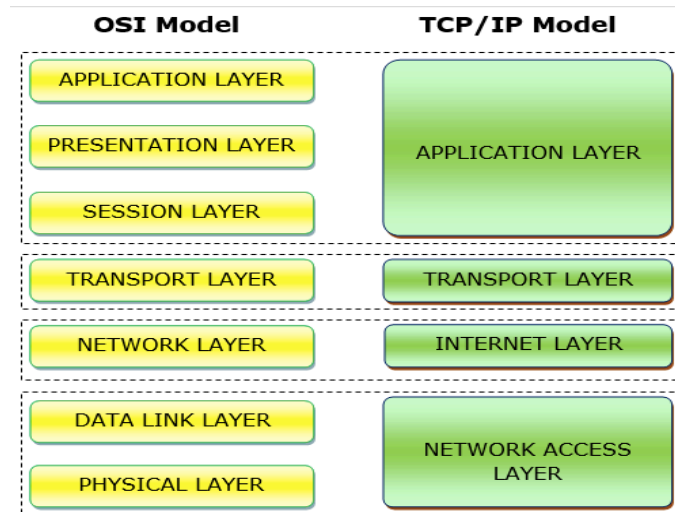
- Selection of a packet switching network which is based on a connectionless internetwork layer is called an internet layer provide all the functionality that network layer provides in OSI model.
- It is the layer which holds the whole architecture together.
- It is responsible for finding the correct path for datagram [packet].
- It helps the packet to travel independently to the destination.
- Order in which packets are received is different from the way they are sent.
- IP (Internet Protocol) is used in this layer.

■ Host-to-network Layer :

- A lowest layer and also called network access layer
- It deals with LAN and WAN protocols and all the functionality provided by physical and data link layer in OSI model.
- Protocol is used to connect to the host, so that the packets can be sent over it.
- Varies from host to host and network to network.

Similarities between OSI and TCP / IP Reference Models

- Both models are based upon layered architecture.
- The layers in the models are compared with each other.



- In both the models, protocols are defined in a layer-wise manner.
- In both models, data is divided into packets and each packet may take the individual route from the source to the destination.

Differences between OSI and TCP / IP

OSI model

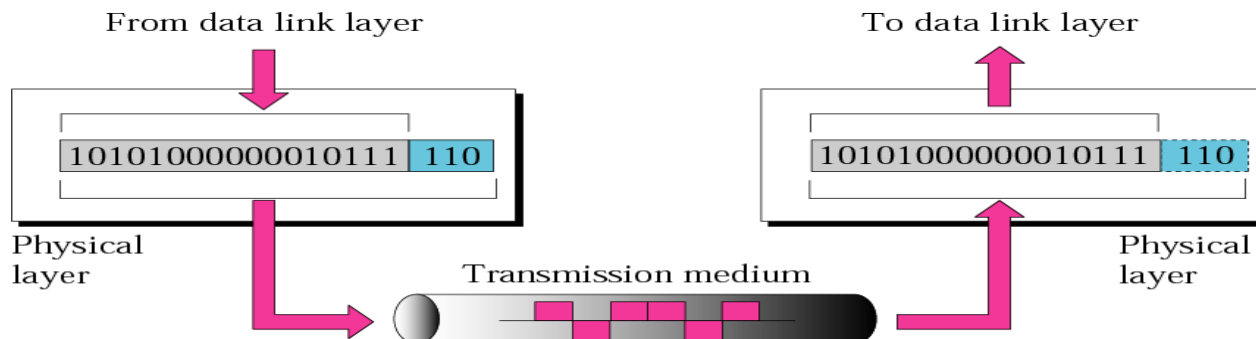
- It is a generic model that is based upon functionalities of each layer.
- distinguishes the three concepts, namely, **services**, **interfaces**, and **protocols**.
Services are set of operations and Interface provides a way to implement them
- gives guidelines on how communication needs to be done
- the model was developed first and then the protocols in each layer were developed.
- The OSI has seven layers while the

TCP/IP

- It is a protocol-oriented standard.
- TCP/IP does not have a clear distinction between these three.
- TCP/IP protocols layout standards on which the Internet was developed. So, TCP/IP is a more practical model.
- the protocols were developed first and then the model was developed.
- TCP/IP has four layers.

Physical Layer

- ✎ The physical layer coordinates the functions required to carry a **bit stream** over a **physical medium**.
- ✎ It deals with the **mechanical and electrical specifications** of the interface and transmission medium.
- ✎ It also **defines the procedures and functions** that physical devices and interfaces have to perform for transmission to occur.



The physical layer is responsible for movements of individual bits from one hop (node) to the next.

Layer 1: The Physical Layer :

 lowest layer of the OSI Model.

- It activates, maintains and deactivates the physical connection.
- It is responsible for transmission and reception of the unstructured raw data
- Voltages and data rates needed for transmission is defined in the physical layer.
- It converts the digital/analog bits into electrical signal or optical signals.
- Data encoding is also done in this layer.

The physical layer is also concerned with the following:

1. **Physical characteristics of interfaces and medium.** The physical layer defines the characteristics of the interface between the devices and the transmission medium. It also defines the type of transmission medium.
2. **Representation of bits.** The physical layer data consists of a stream of bits (sequence of 0s or 1s) with no **interpretation**. To be transmitted, bits must be **encoded into signals - electrical or optical**. The physical layer defines the type of encoding (how 0s and 1s are changed to signals).
3. **Data rate.** The transmission rate - the number of bits sent each second - is also defined by the physical layer. In other words, the physical layer defines the duration of a bit, which is how long it lasts.
4. **Synchronization of bits.** The sender and receiver not only must use the same bit rate but also must be synchronized at the bit level. In other words, the sender and the receiver clocks must be synchronized.

Contd.

5. **Line configuration.** The physical layer is concerned with the connection of devices to the media (point – to – point or multipoint).
6. **Physical topology.** The physical topology defines how devices are connected to make a network.
7. **Transmission mode.** The physical layer also defines the direction of transmission between two devices: **simplex, half-duplex, or full-duplex.**

Data Representation

📢 The word ***data*** refers to information presented in whatever form is agreed upon by the parties creating and using the data.

📢 ***Signal:*** the electrical wave that is used to represent the data.

📢 It Can be analog or digital signal

Data Representation Techniques

- 📢 The type of data to be transmitted can be in the form of **text, audio, and video** in the form of electrical signal, radio, laser, or other radiated energy source.
- 📢 Data can be analog or digital
- 📢 The term **analog** data refers to information that is continuous;
- 📢 **digital** data refers to information that has discrete states.
- 📢 Example: **analog clock** (with second, minute and hour hands) and digital clock

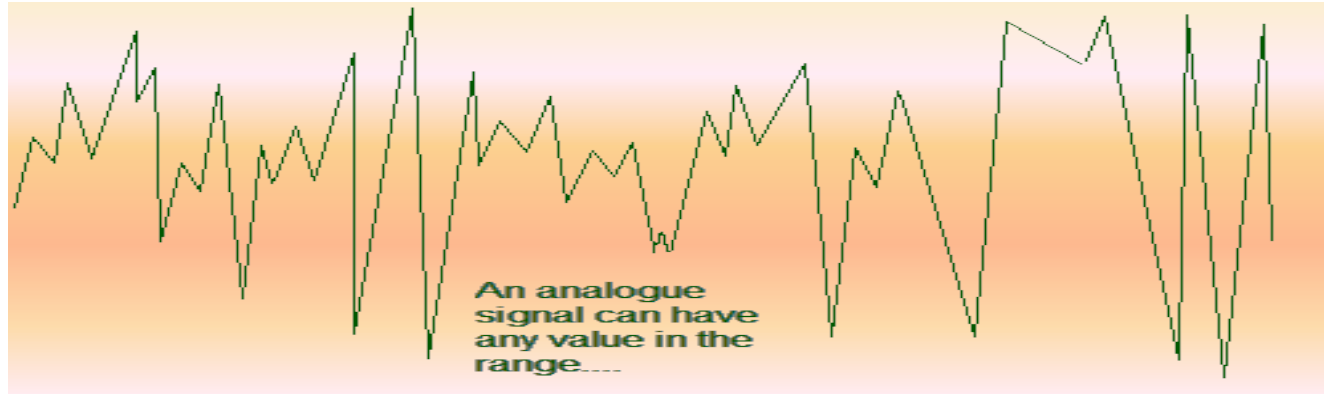
Analog and digital signals

- 🔊 An analog signal has infinitely **many levels of intensity** over a period of time.
 - 🔊 As the wave moves from value A to value B, it passes through and includes an infinite number of values along its path.
- 🔊 A digital signal, on the other hand, can have only a **limited number** of defined values.
 - 🔊 Although each value can be any number, it is often as simple as 1 and 0.

Analog Signals

- 📢 Are Continuous wave that carries information by altering the characteristics of waves.
- 📢 Analogue means that the original information is retransmitted to the receiver/listener **without any manipulation**.
- 📢 Here the signal can take on **any value** (within the limits set by the recording equipment and the transmitter).
- 📢 It measures rather than counts
- 📢 For instance, Voice and all sounds are analog, traveling to human ears in the forms of waves.

Contd.



- 📢 Radio, telephone and recording equipment historically have been analog, but they are beginning to change –due to large to computers to the other types of signals-digital
- 📢 **The disadvantage of analogue signals** is that any 'noise', interference, added to the signal at any point cannot be removed from the audio signal and this degrades the audio quality of the signal or causes 'hiss'.

Digital Signals

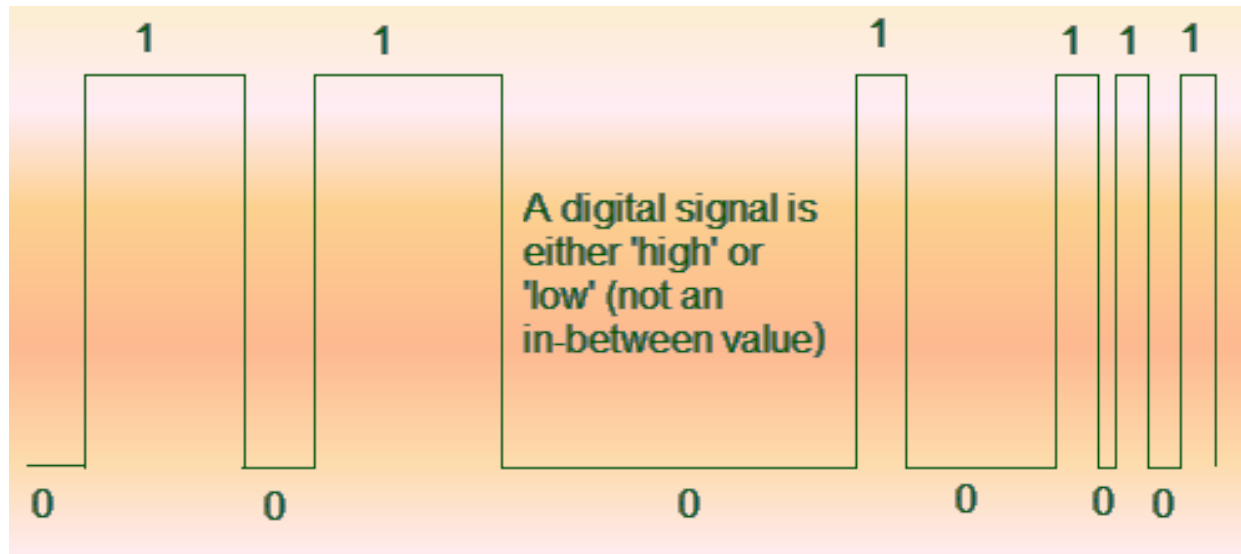
- 📢 are a discrete series of pulses - either **high or low - on or off** - sometimes expressed as binary code 1s and 0s.
- 📢 It counts but not measures
- 📢 Discrete pluses of data transmission rather than continues wave
- 📢 More prevalent in computer based devices
- 📢 **Flashing light and telegraph message** are traditional examples of on-off pluses, although not in binary code.

Contd.

The advantage of digital signals is that

1. they are much less likely to be degraded by interference (noise).
2. also it is possible to send a lot more information digitally (e.g. more television channels) than using analogue technology.
3. They can be transmitted effectively along fibre optic cables. (on - light signal, off - no light signal) and these are cheaper to manufacture than copper wiring and can cope with multiple signals without interference between them.
4. They can be sent directly to computers which use digital systems themselves.

Contd.



📢 The signals from a satellite are digital. They use a system called 'packet switching' which is the same as used in computer networks and for the internet. This system is particularly effective for dealing with noise elimination.

Contd.

Which of the following signals are Analog and which are Digital?

- 🔊 Volume control on a radio
- 🔊 Traffic lights
- 🔊 Motor bike throttle
- 🔊 Water tap
- 🔊 Dimmer switch
- 🔊 Light switch
- 🔊 In Music on a CD
- 🔊 Music on a tape



Reading Assignment

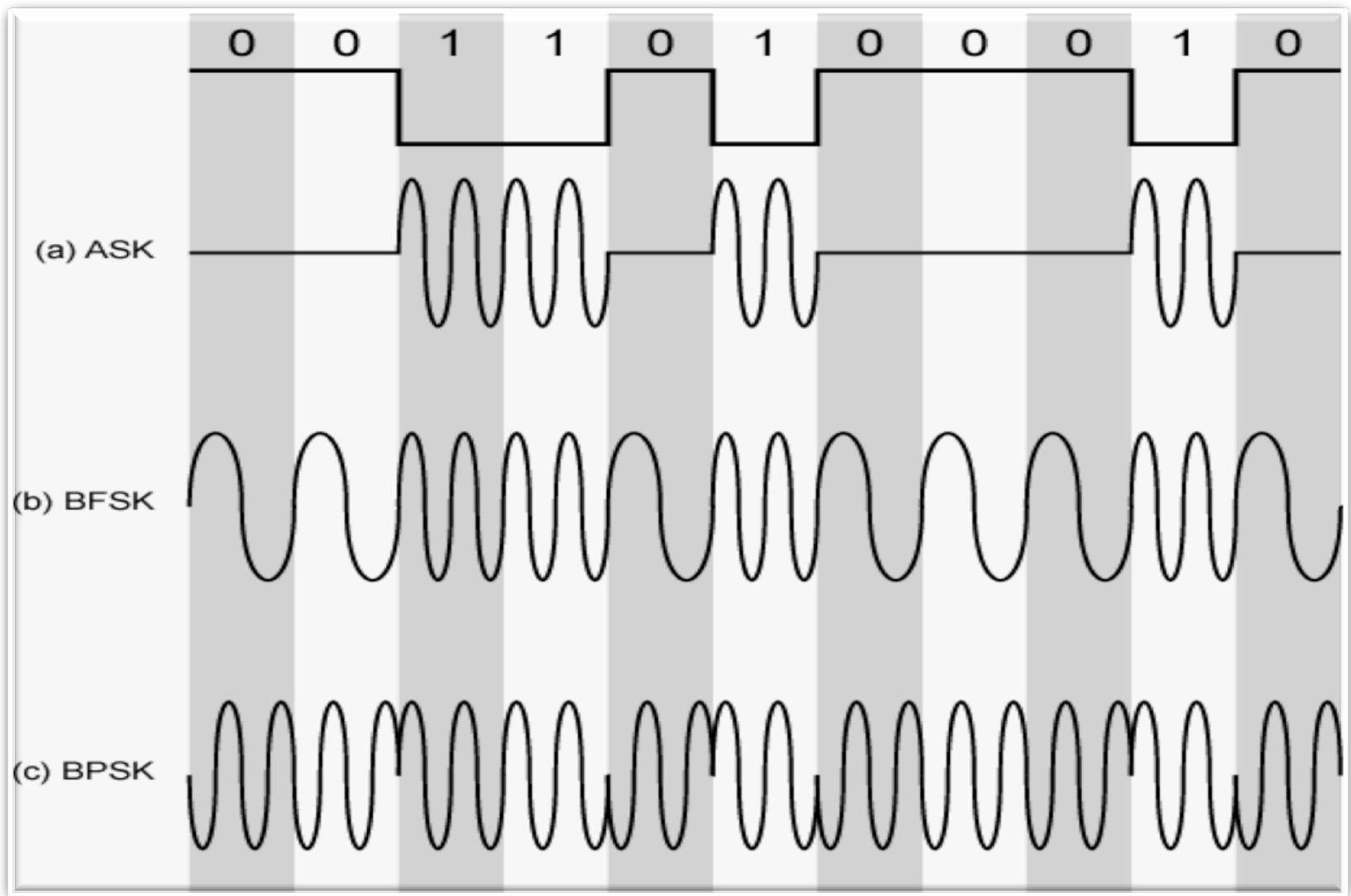
MODULATION /DATA ENCODING

Modulation

- **Modulation** is the process of transforming a carrier signal so that it can carry the information of a message signal.
- It superimposes the contents of the message signal over a high-frequency carrier signal, which is then transmitted over communication channels.
- Modulation can be of two types –
 - Analog Modulation
 - Digital Modulation

Digital Data to Analog Signal Encoding Techniques

- Modulation involves operation on one or more of the three characteristics of a carrier signal: **amplitude, frequency, and phase.**
- Accordingly, there are three basic encoding or modulation techniques for transforming digital data into analog signals
 1. Amplitude-shift keying (ASK)
 2. Frequency-shift keying (FSK)
 3. Phase-shift keying (PSK)

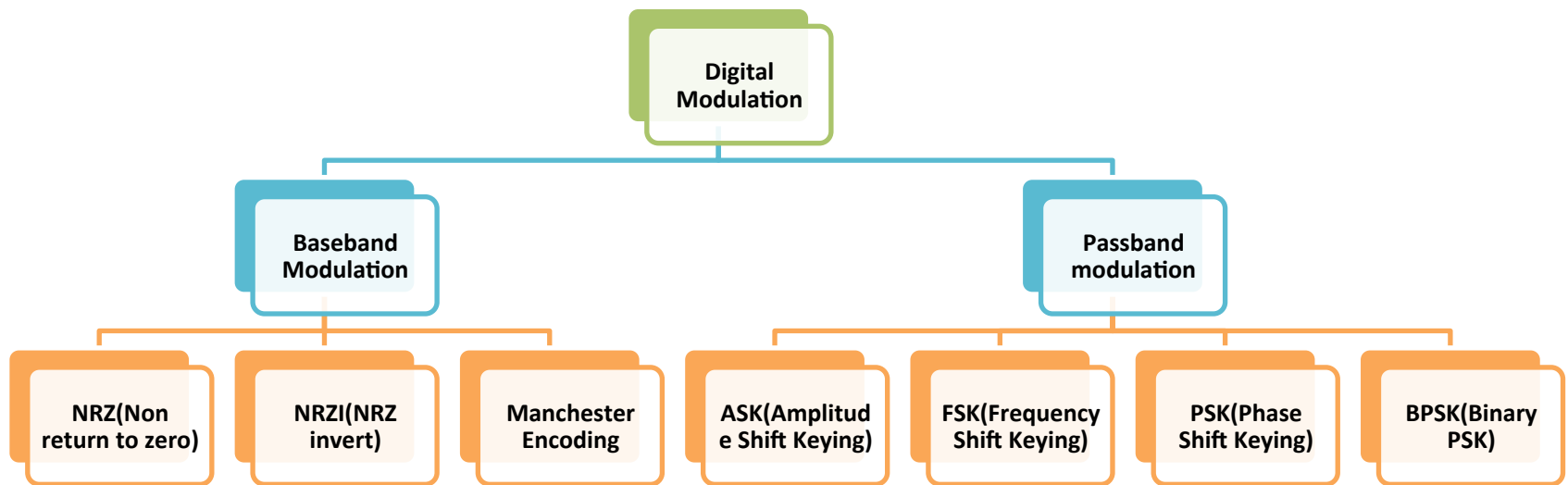


Analog Modulation

- Here, the analog information signal is transformed to the analog carrier signal so that it can travel large distances without substantial loss.
- Analog modulation can be of three types –
 - Amplitude Modulation
 - Frequency Modulation
 - Phase Modulation

Digital Modulation

- Digital modulation is the process of converting a digital bit stream into an analog carrier wave for transmission via a communication channel.
- Digital modulation is broadly divided into two categories –
 - Bandpass Modulation as in baseband transmission:**
 - Here, the bits are converted directly into signals.
 - Passband Modulation as in passband transmission:**
 - Here, the amplitude, phase or frequency of the carrier signal is regulated to transmit the bits.



DIGITAL TRANSMISSION:

- how to convert digital data into digital signals ?
- It can be done in two ways:
 - line coding
 - block coding.
- For all communications, line coding is necessary whereas block coding is optional.

Digital-to-Digital Conversion

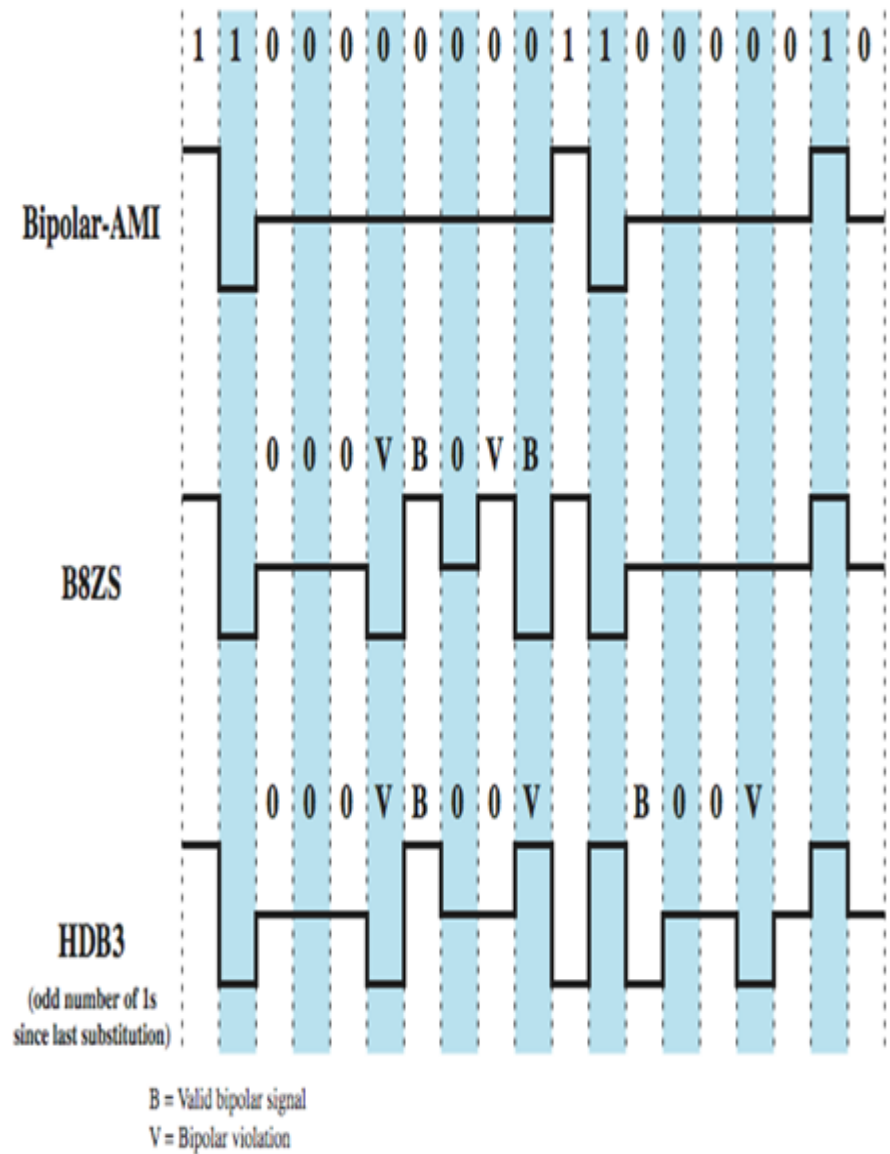
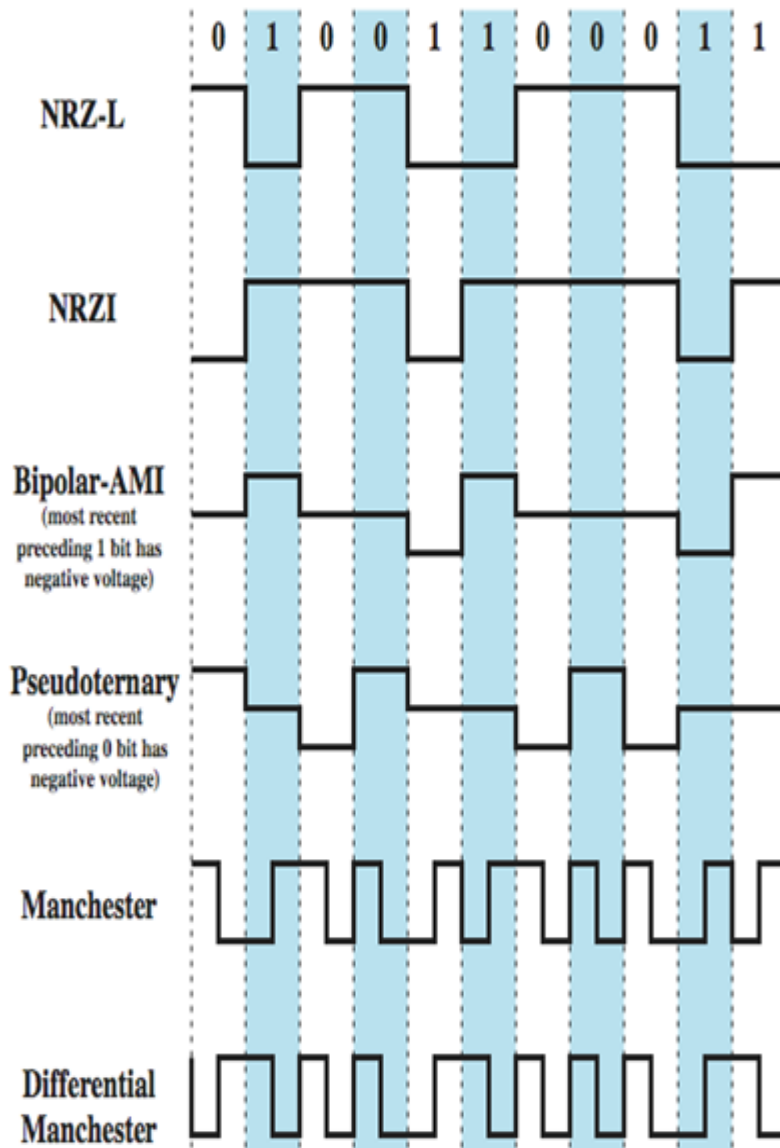
- **line coding**
 - **Uni-polar**
 - **Bipolar**
 - **Polar** :4 types
 - Polar Non Return to Zero (Polar NRZ)
 - Return to Zero (RZ)
 - Manchester
 - Differential Manchester
- **block coding.**
 - involves three steps:
 - Division
 - Substitution
 - Combination.

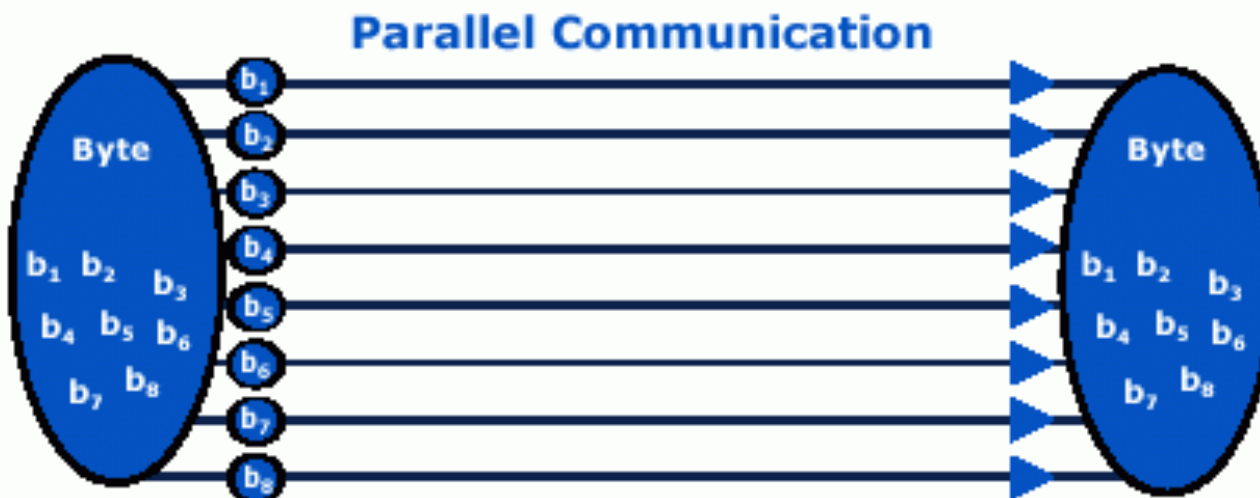
Analog-to-Digital Conversion

- Analog data is a continuous stream of data in the wave form whereas digital data is discrete. To convert analog wave into digital data, we use Pulse Code Modulation (PCM).
- PCM is one of the most commonly used method to convert analog data into digital form. It involves three steps:
 - Sampling
 - Quantization
 - Encoding.

Assignment #1

- Describe the following digital signal encoding formats (which format is used by Ethernet?)
 1. **Non return-to-Zero-Level (NRZ-L)**
 2. **Non return to Zero Inverted (NRZI)**
 3. **Bipolar-AMI**
 4. **Pseudoternary**
 5. **Manchester**
 6. **Differential Manchester**





Serial transmission can be either asynchronous or synchronous.

■ Asynchronous Serial Transmission

- It is named so because there is no importance of timing. Data-bits have specific pattern and they help receiver recognize the start and end data bits. For example, a 0 is prefixed on every data byte and one or more 1s are added at the end.
- Two continuous data-frames (bytes) may have a gap between them.

■ Synchronous Serial Transmission

- Timing in synchronous transmission has importance as there is no mechanism followed to recognize start and end data bits. There is no pattern or prefix/suffix method. Data bits are sent in burst mode without maintaining gap between bytes (8- bits). Single burst of data bits may contain a number of bytes. Therefore, timing becomes very important.
- It is up to the receiver to recognize and separate bits into bytes. The advantage of synchronous transmission is high speed, and it has no overhead of extra header and footer bits as in asynchronous transmission.

ANALOG TRANSMISSION

- To send the digital data over an analog media, it needs to be converted into analog signal. There can be two cases according to data formatting.
- **Bandpass:** The filters are used to filter and pass frequencies of interest. A bandpass is a band of frequencies which can pass the filter.
- **Low-pass:** Low-pass is a filter that passes low frequencies signals.
- When digital data is converted into a bandpass analog signal, it is called digital-to- analog conversion. When low-pass analog signal is converted into bandpass analog signal, it is called analog-to-analog conversion.

Digital-to-Analog Conversion

- When data from one computer is sent to another via some analog carrier, it is first converted into analog signals. Analog signals are modified to reflect digital data.
- An analog signal is characterized by its amplitude, frequency, and phase.
- There are three kinds of digital-to-analog conversions:
 - Amplitude Shift Keying
 - Frequency Shift Keying
 - Phase Shift Keying

Analog-to-Analog Conversion

- Analog signals are modified to represent analog data. This conversion is also known as **Analog Modulation**.
- Analog modulation is required when bandpass is used.
- Analog to analog conversion can be done in three ways:
 - Amplitude modulation
 - Frequency modulation
 - Phase modulation